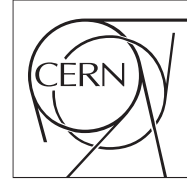


The Compact Muon Solenoid Experiment

Analysis Note

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Measurement of $B_c^+ \rightarrow J/\psi \pi^+$ and $B^+ \rightarrow J/\psi K^+$ production cross sections in pp collisions at $\sqrt{s} = 7$ TeV

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Abstract

We describe the measurements of the differential cross sections of $B_c^+ \rightarrow J/\psi \pi^+$ and $B^+ \rightarrow J/\psi K^+$ processes in pp collisions at $\sqrt{s} = 7$ TeV. The measurements are made with CMS data collected in the 2011 run, corresponding to an integrated luminosity of 4.77 fb^{-1} . The B_c^+ and B^+ differential cross sections are reported as a function of $B_c^+(B^+)$ transverse momentum p_T and absolute rapidity $|y|$. The integrated production cross sections for B_c^+ and B^+ with $p_T(B_c^+, B^+) > 10 \text{ GeV}/c$ and $|y(B_c^+, B^+)| < 1.5$ are measured to be $45.17 \pm 5.25^{+3.00}_{-3.12} \text{ pb}$ and $5919.02 \pm 82.33 \pm 426.17 \text{ pb}$ respectively. The first uncertainties are statistical uncertainties of the data and MC samples and second uncertainties are systematical uncertainties.

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1 Introduction

In the standard model, B^+ ($u\bar{b}$) is the b quark meson which has the largest production rate in hadron collision. The B_c^+ meson contains two heavy quarks of different flavors, $\bar{b}$ and c quarks. The production of B_c^+ is much rare than B^+ production in hadron collision because it need the simultaneous production of $b\bar{b}$ and $c\bar{c}$ pairs. The study of heavy quarks production in high energy hadronic interactions plays a critical role in testing next-to-leading order (NLO) Quantum Chromodynamics (QCD) calculations [1, 2] and more recent predictions by fixed order plus next-to-leading-logarithms (FONLL) [3, 4]. However, the dependence of the theoretical predictions on the renormalization and factorization scales and the b -quark mass m_b results in theoretical uncertainties of 40%. Effective models inspired by quantum chromodynamics [5] have been calculated for B_c meson for two decades. The complete order- α^4 approach [6–15] (α^4 is the strong-interaction coupling) predicts B_c^+ production cross section to be 0.2% of the inclusive $b\bar{b}$ cross section.

The first heavy quark production measurements were performed by UA1 collaboration at $Spp\bar{S}$ (CERN) at center-of-mass energy of $\sqrt{s} = 630$ GeV [16, 17]. Then CDF and D0 collaborations at the Fermilab Tevatron measured more heavy quark productions at $\sqrt{s} = 1.8$ TeV and 1.96 TeV [18–25] and observed the B_c in the semileptonic channel $B_c \rightarrow J/\psi l \nu$ and hadronic channel $B_c \rightarrow J/\psi \pi$ [26–29]. After 2010, experiments at LHC achieved substantial progress in the understanding of heavy quark productions. LHCb collaboration measured b hadrons, B_c absolute and relative production cross sections in the forward rapidity region at $\sqrt{s} = 7$ TeV [30–33]. CMS collaboration measured production cross sections for B^+ , B^0 , B_s , Λ_b , inclusive b hadrons production and relative B_c productions [34–41]. ATLAS measured B^+ and b hadrons production cross sections [42–44]. However, only relative production ratio of B_c^+ to B^+ or B_s^0 cross sections have been measured at LHCb and CMS. There is not yet absolute production cross section measurement for any B_c meson decay channel. The production cross sections of B^+ production measurement [44] shows different transverse momentum and rapidity dependencies compared with MCNLO+HERWIG prediction. Therefore, a precise measurements of B^+ and B_c^+ absolute productions as a function of transverse momentum and rapidity at CMS will provide useful information on the unknown production mechanism of B^+ and B_c^+ mesons and is also important to guide b quark meson studies at the LHC.

In this paper, we report on the first absolute production cross section measurement of $B_c^+ \rightarrow J/\psi \pi^+$ and the production cross section of $B^+ \rightarrow J/\psi K^+$ processes at CMS experiment [45]. We use a sample of pp collision data at 7 TeV corresponding to an integrated luminosity of 4.77 fb^{-1} recorded by the CMS experiment. The differential cross sections are reported with respect to B_c^+ (B^+) p_T and absolute rapidity $|y|^{*)}$. Using the large $B^+ \rightarrow J/\psi K^+$ sample, differential cross section of $d\sigma/dp_T(B^+)$ for small and large rapidity events are investigated separately. The inclusion of charge conjugate modes is implied throughout this paper. The results are compared to theoretical predictions based on QCD.

2 Data and Monte Carlo Samples

This analysis has been performed with 4.77 fb^{-1} data recorded by CMS detector in 2011 LHC run at $\sqrt{s} = 7$ TeV. The data include CMS 2011A and 2011B datasets corresponding to LHC running with different instantaneous luminosities in 2011, the cross section measurements are systematically limited with the data samples used for this analysis. The datasets run range is from run 165071 to 180252, summarized in Table 1. Run numbers are consistent with the CMS T&P single muon efficiencies studies [46].

Good runs and luminosity sections are selected with the Muon Physics JSON file [47]. It guarantees the reliability of the data, especially for the high quality running of CMS muons and tracker sub-detectors.

*) Rapidity y is defined as $\frac{1}{2} \ln \frac{E+p_L}{E-p_L}$, E is the particle energy, p_L is the particle momentum along the counter clockwise beam direction.

Table 1: Datasets used in the analysis

Sample	Run range	Number of events	$\mathcal{L}(\text{pb}^{-1})$	trigger menu
/MuOnia/Run2011A-PromptReco-v4/AOD	165071-168437	26987206	970	1E33+1.4E33
/MuOnia/Run2011A-PromptReco-v5/AOD	170053-172619	8202804	357	2E33
/MuOnia/Run2011A-PromptReco-v6/AOD	172620-175770	9501198	706	2E32+3E33
/MuOnia/Run2011B-PromptReco-v1/AOD	175832-180296	27752865	2741	3E33+5E33

• Cert_160404-180252_7TeV_PromptReco_Collisions11_JSON_MuonPhys.txt

The data luminosity $4.77 \times (1 \pm 4\%) \text{ fb}^{-1}$, is computed with the officially recommended tools from the luminosity group [48]. We have performed the analysis in the package of CMSSW_4_4_5. The Global Tag is GR_P_V27 for /MuOnia/Run2011A-PromptReco/AOD and /MuOnia/Run2011B-PromptReco/AOD.

To validate the analysis technique, we also performed detailed studies based on Monte Carlo (MC) samples, listed in Table 2. MC samples are also used to determine portions of the reconstruction efficiencies and variety of systematic checks.

Table 2: Monte Carlo Samples used in the analysis

Sample	Events	$\varepsilon_{\text{filter}}$	$\sigma(\text{pb})$	$\mathcal{L}(\text{pb}^{-1})$
/BctoJPsiPi_Bcvepy_Sum12/xywang-BctoJPsiPi_HLTBPH2011_RECO-0a2dd225eed315a6c242ad2b584c6d9/USER	1490439		301162	
/BpToPsiMuMu_2MuPEtaFilter_Tight_7TeV-pythia6-evtgen/Fall11-HLTBPh2011_START42_V14B-v2/AODSIM	8380592	3.188×10^{-4}	2.98×10^7	842
/B0ToPsiMuMu_2MuPEtaFilter_Tight_7TeV-pythia6-evtgen/Fall11-HLTBPh2011_START42_V14B-v2/AODSIM	7221225	3.114×10^{-4}	2.65×10^7	969
/BsToPsiMuMu_2MuPEtaFilter_Tight_7TeV-pythia6-evtgen/Fall11-HLTBPh2011_START42_V14B-v2/AODSIM	2092273	1.004×10^{-4}	2.41×10^7	826
/LambdaBToPsiMuMu_2MuPEtaFilter_Tight_7TeV-pythia6-evtgen/Fall11-HLTBPh2011_START42_V14B-v2/AODSIM	2184854	6.4×10^{-5}	8000000	3906
/JPsiToMuMu_2MuPEtaFilter_7TeV-pythia6-evtgen/Fall11-HLTBPh2011_START42_V14B-v2/AODSIM	5545894	0.04923	8029000	133

Because of very low PYTHIA B_c production rate ($\sim 10^{-6}$) in a minimum bias generation, we used the MC samples of $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ process generated by the BCVEGPY generator in the analysis. The pa-

parameters defined during the generation by BCVEGPY includes the mass of B_c ($m(B_c) = 6.275 \text{ GeV}/c^2$), mass of b quark ($m(b) = 5.000 \text{ GeV}/c^2$) and mass of c quark ($m(c) = 1.275 \text{ GeV}/c^2$). The other parameters include $R(0)$ set to 1.241 and $R'(0)$ is 0.44833 where $R(0)$ is the radial wave function ($\text{GeV}^{3/2}$) and $R'(0)$ is first derivative of the radial wave function ($\text{GeV}^{5/2}$). During the PYTHIA generation the used parameters are PDF(CTEQL1) and MSEL is 1.

In the generation, unweighted Les Houches Event (LHE) files including the information of B_c and two spectator quarks was generated by BCVEGPY [49, 50] and then these files are exported to CMSSW. The B_c decay, hadronization of two spectator quarks, ISR and FSR are simulated by PYTHIA in CMSSW. The B_c meson was generated in the phase space of B_c transverse momentum $p_T > 8 \text{ GeV}/c$ and absolute rapidity $|y| < 2.5$. In the simulation configuration file, the kinematic filter requirements of J/ψ 's daughter muons are pseudo-rapidity $|\eta| < 2.5$ and $p_T > 2.5 \text{ GeV}/c$. The $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ sample are generated by PYTHIA, the kinematic filter requirements of muons are same as in $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ generation. Samples are published in DBS and available with the datasets names listed in Table 1. The configuration files are generated by cmsDriver commands and can be found in V00-00-00 of UserCode/xywang/Bcconfig. To determine the detector acceptance for the signal process, MC samples of $pp \rightarrow B_c + X$, $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ and $pp \rightarrow B^\pm + X$, $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ processes without any kinematic requirement were generated.

2.1 PU reweighting for MC Samples

The entire machinery for pileup reweighting is documented on the web page https://twiki.cern.ch/twiki/bin/viewauth/CMS/Pileup_2011_Reweighting. Instructions as for the Fall11 samples have been followed. Figure 1 shows number of Primary Vertices distributions for B_c and $B^\pm$ for data and MC before and after PU reweighting.

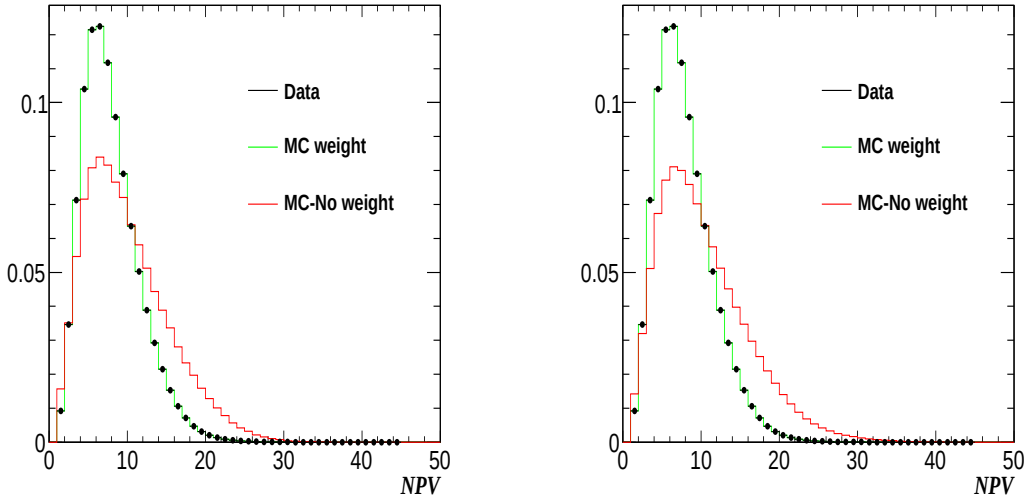


Figure 1: Primary Vertex distributions for data compared to B_c (left) and $B^\pm$ (right) simulation with and without PU reweighting.

3 Event Reconstruction

The process $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ reconstruction contains the reconstruction of $J/\psi(\mu^+\mu^-)$ and the reconstruction of $B_c \rightarrow J/\psi\pi$ process. The J/ψ reconstruction is defined by the dimuon Displaced trigger cuts and the reconstruction of $B_c \rightarrow J/\psi\pi$ process includes a kinematic fit the $J/\psi(\mu^+\mu^-)$ and a π track. The same analysis strategy and selection criteria to reconstruct $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ and $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ process has been used .

3.1 Trigger paths and J/ψ reconstruction

The J/ψ is reconstructed by dimuon decay, each muon is accepted by CMS detector. The datasets were pre-selected by the CMS muon displaced trigger with the evolution of trigger menu in 2011 data taking. In the muon displaced trigger, dimuon vertex is required to be displaced from the beamspot in order to reject prompt J/ψ background.

Table 3 shows trigger paths during the 2011 data taking evolved with the increase of the LHC instantaneous luminosities. Table 4 shows various displaced J/ψ trigger criteria with increasingly tight thresholds throughout 2011. Main selections criteria focus on the J/ψ and muon p_T requirements, probability of two muons vertex constraint kinematic fit and decay length of two muon. Variables used in the trigger path are explained in the followings.

- $\cos \alpha$: The angle in xy plane between the J/ψ momentum vector and position vector from the beamspot to the reconstructed dimuon vertex
- L_{xy}/σ_{xy} : L_{xy} is the transverse separation between the dimuon vertex and the beamspot, σ_{xy} is uncertainty on L_{xy}
- VP : The probability of the kinematic fit of two muons sharing one vertex
- DCA : Distance of closest approach between two muons.

Table 3: HLT trigger menu for dimuon evolved with the LHC luminosity increase in 2011.

Run range	Menu	Trigger path	$\mathcal{L}(\text{pb}^{-1})$	L1 seed
165088-167043	1e33	HLT_Dimuon7_Jpsi_Displaced_v1	709.0	L1_DoubleMu0
166346-166346	1e33	HLT_Dimuon7_Jpsi_Displaced_v2	4.3	L1_DoubleMu0
167078-167913	1.4e3	HLT_Dimuon7_Jpsi_Displaced_v3	244.7	L1_DoubleMu0
170249-172868	2e33	HLT_DoubleMu3p5_Jpsi_displaced_v2	824.5	L1_DoubleMu0
173236-177053	3e33	HLT_DoubleMu4_Jpsi_displaced_v1	2071	L1_DoubleMu0_HighQ
178421-179889	5e33	HLT_DoubleMu4_Jpsi_displaced_v4	697.4	L1_DoubleMu0_HighQ
179959-180252	5e33	HLT_DoubleMu4_Jpsi_displaced_v5	113.0	L1_DoubleMu0_HighQ

For events selection, muons are required to fulfill one of the acceptance criteria listed below .

- $p_T^\mu > 3.5\text{GeV}/c$ and $|\eta^\mu| < 1.2$ or
- $2\text{GeV}/c < p_T^\mu < 3.75\text{GeV}/c$ and $1.2 < |\eta^\mu| < 1.6$; or
- $p_T^\mu > 2\text{GeV}/c$ and $1.6 < |\eta^\mu| < 2.4$.

Table 4: Criteria of displaced trigger paths.

Trigger path	Different selection	Common selection
HLT_Dimuon6p5_Jpsi_Displaced_v1	$p_T^{J/\psi} > 6.5 \text{ GeV}/c$ $VP > 0.005, \eta_\mu < 2.5$	
HLT_Dimuon7_Jpsi_Displaced_v1(v2,v3)	$p_T^{J/\psi} > 6.9 \text{ GeV}/c$ $VP > 0.005, \eta_\mu < 2.5$	$\cos \alpha > 0.9,$ $L_{xy}/\sigma_{xy} > 3,$
HLT_DoubleMu3p5_Jpsi_displaced_v2	$p_T^{J/\psi} > 6.9, VP > 0.1$ $ \eta_\mu < 2.2, p_T^\mu > 3.5 \text{ GeV}/c$	$M_{J/\psi}(2.9,3.3),$ $DCA < 0.5$
HLT_DoubleMu4_Jpsi_displaced_v1(v4,v5)	$p_T^{J/\psi} > 6.9 \text{ GeV}/c, VP > 0.15$ $ \eta_\mu < 2.2, p_T^\mu > 4 \text{ GeV}/c$	

Both displaced trigger selected muons decayed from J/ψ are required to be either Tracker muon or Global muon categories. Then we require that Tracker muon and Global muon objects pass the arbitration algorithm as the followings. The muon acceptance criteria and muon arbitration algorithm are same as CMS approved muon T-P analysis [46, 51].

- μ track normalized $\chi^2 < 1.8$
- Number of hits in pixel > 1
- Number of hits in tracker > 10
- Muon $|dB| < 3.0 \text{ cm}$ and $dz < 15 \text{ cm}$
- Pass the TrackerMuonArbitrated and TMOneStationTight selections
- Global track $\chi^2/ndof < 20$ and number of valid hit of global track > 0 , only for the global muons.

After above listed pre-selections by the displaced trigger with the evolution of trigger menu, we use the selection criteria of trigger path HLT_DoubleMu4_Jpsi_displaced_v4 in the code for the final $J/\psi \rightarrow \mu^+\mu^-$ selection, listed below

- Opposite charge pair of muons and $\cos \alpha > 0.9$
- $L_{xy}/\sigma_{L_{xy}} > 3$ to select long life time J/ψ 's
- $p_T^\mu > 4 \text{ GeV}/c$ and $\eta^\mu < 2.4$
- $p_T^{J/\psi} > 6.9 \text{ GeV}/c$
- $DCA < 0.5 \text{ cm}$
- Probability of two-muon Kalman-Vertex Fit $> 15\%$

The above J/ψ selections criteria cover the trigger criteria of the all the trigger paths in 2011. Therefore, the trigger efficiency in the analysis can be applied for the full 2011 datasets without any luminosity rescaling. We keep all the J/ψ candidates that have an invariant mass within $\pm 120 \text{ MeV}/c^2$ to the PDG value [52]. Figure 2 shows the invariant mass distribution of two muons for 2011 data after passing above selections criteria.

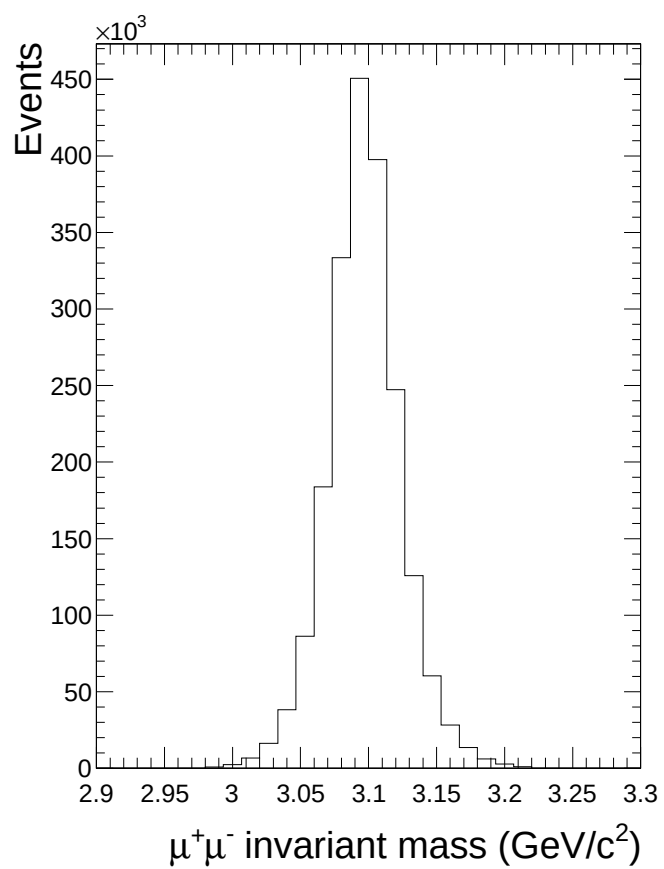


Figure 2: J/ψ signal in the invariant mass distribution of two muons for 2011 data.

3.2 Reconstruction of $B_c^\pm \rightarrow J/\psi \pi^\pm$ and $B^\pm \rightarrow J/\psi K^\pm$

To reconstruct B_c and $B^\pm$ meson, the selected J/ψ candidate is combined with a pion or kaon track, fit the two muons decayed from J/ψ candidate and a track to a common vertex by mass constrained kinematic vertex fit. The tracks are required to pass the following cuts.

- $\pi (K)$ track fit normalized $\chi^2 < 3$
- $\pi (K)$ track number of hits in pixel > 1
- $p_T^{\pi (K)} > 2.3 \text{ GeV}/c$
- $\pi (K)$ number of hits in tracker > 10 .

For B_c selections, pions and J/ψ candidates that fulfill all the requirements are subjected to the combined vertex and kinematic fit (Kinematic Constrained Vertex Fitter), in which the mass of dimuon is constrained to be the nominal J/ψ mass [52] and tracks are considered to be pion particles. B_c selection cuts have been optimized by maximizing the observed B_c events, ratio of signal and background (S/N), significance ($CL = S/\sqrt{S+B}$), within 2σ of B_c mass spectrum in data and MC samples. The obtained cuts are listed below

- The probability of the combined vertex and kinematic fit > 0.03
- $\cos\theta > 0.98$, where θ is the angle in the xy plane between the B_c momentum vector and position vector from the primary vertex to the reconstructed second vertex
- The B_c decay length $L_{xy}/\sigma_{L_{xy}} > 4.0$, this cut is to depress the short lifetime backgrounds where the transverse flight length L_{xy} is the projection of the vector $\vec{s}$ pointing from the primary to the secondary vertex onto the transverse momentum and $\sigma_{L_{xy}}$ is the uncertainty on L_{xy} .

$$L_{xy} = \frac{\vec{s} \cdot \vec{p}_T^{B_c}}{|\vec{p}_T^{B_c}|} \quad (1)$$

Finally, B_c candidates are required to pass transverse momentum $p_T > 10 \text{ GeV}/c$ and absolute rapidity $|y| < 1.5$ cuts. The same selections are applied for the reconstruction of $B^\pm \rightarrow J/\psi K^\pm$ process. Selections criteria above base on similar studies as in BPH-12-011 analysis [41]. We also performed comparative studies on selection of all $B_c(B^\pm)$ candidates passing full selections and selecting only one $B_c(B^+)$ candidate with maximum p_T in a single event. The difference in signal yields are found to be 2.9% for B_c and 3.7% for $B^\pm$. Due to small difference in yield, we decided to keep all $B_c(B^\pm)$ candidates passing full selections. Figure 3 shows the p_T and $|\eta|$ distributions of muons, J/ψ , pions and kaons after all the $B_c (B^\pm)$ selection cuts.

Figure 4 shows the J/ψ and a pion track invariant mass distribution for 2011 CMS data after applying all the selection criteria. Clear B_c peak can be seen, significance ($CL = S/\sqrt{S+B}$ in two sigma of the B_c mass spectrum) of the B_c signal is 11.4. The observed B_c events are 309 ± 35 . The mass of B_c is fitted to be $6.2725 \pm 0.0029 \text{ GeV}/c^2$, consistent with the PDG average value [52]. The Figure 5 shows the J/ψ and a kaon track invariant mass distribution. The enhancement in the left of $B^\pm$ peak is mainly the $B^0 \rightarrow J/\psi K^{*0}(K^\pm \pi^\mp)$ background with a missing pion or kaon track [34]. In the right of $B^\pm$ peak, the contribution of $B^\pm \rightarrow J/\psi \pi^\pm$ process is fixed to the parameters obtained from the MC simulation. The observed $B^\pm$ events are fitted to be 117486 ± 359 . The mass of $B^\pm$ is $5.27836 \pm 5.92142 \times 10^{-05} \text{ GeV}/c^2$, consistent with the PDG average value [52].

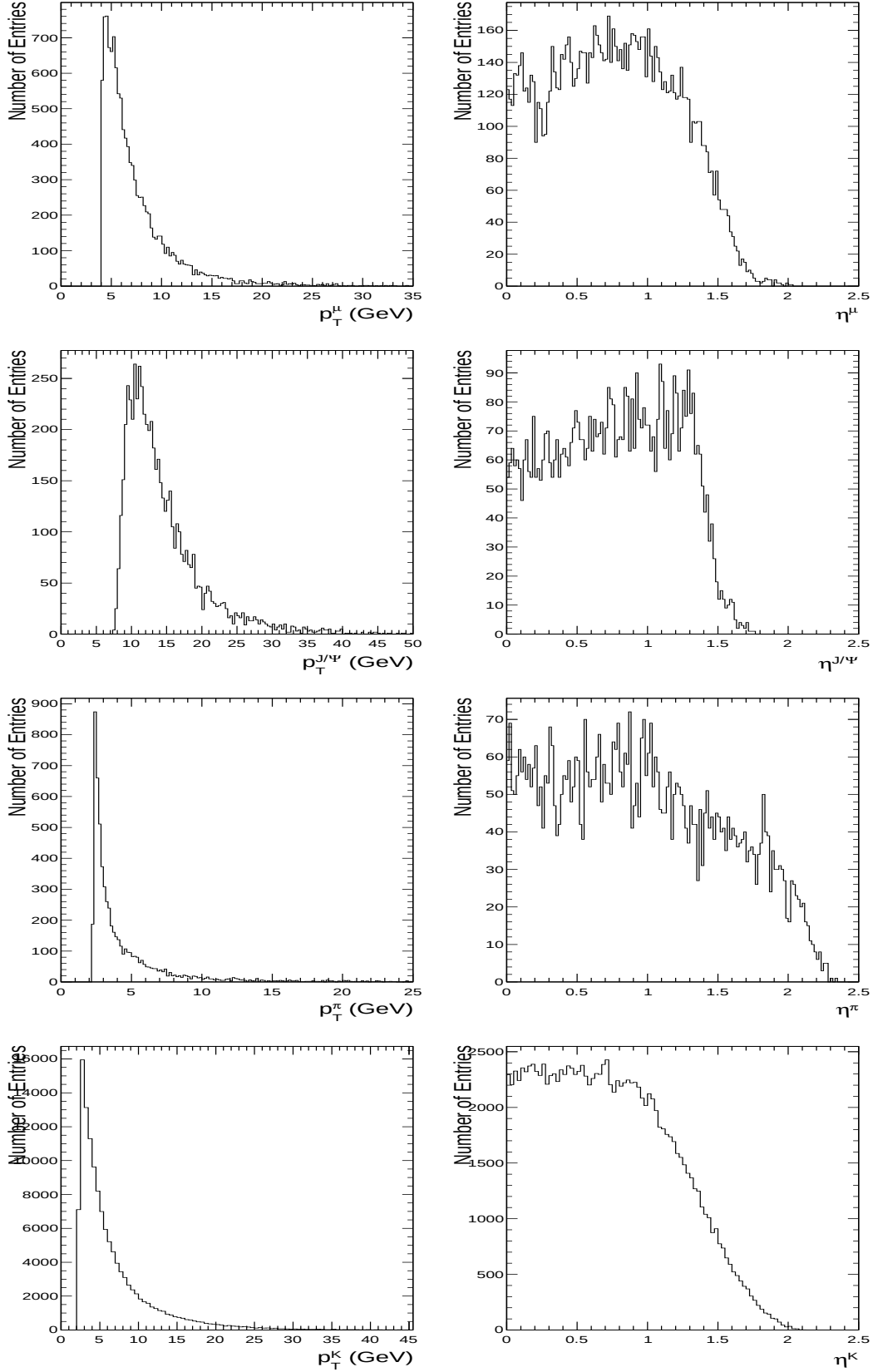


Figure 3: Distributions of p_T and $|\eta|$ of muons, J/ψ 's, pions and kaons after final selections in data.

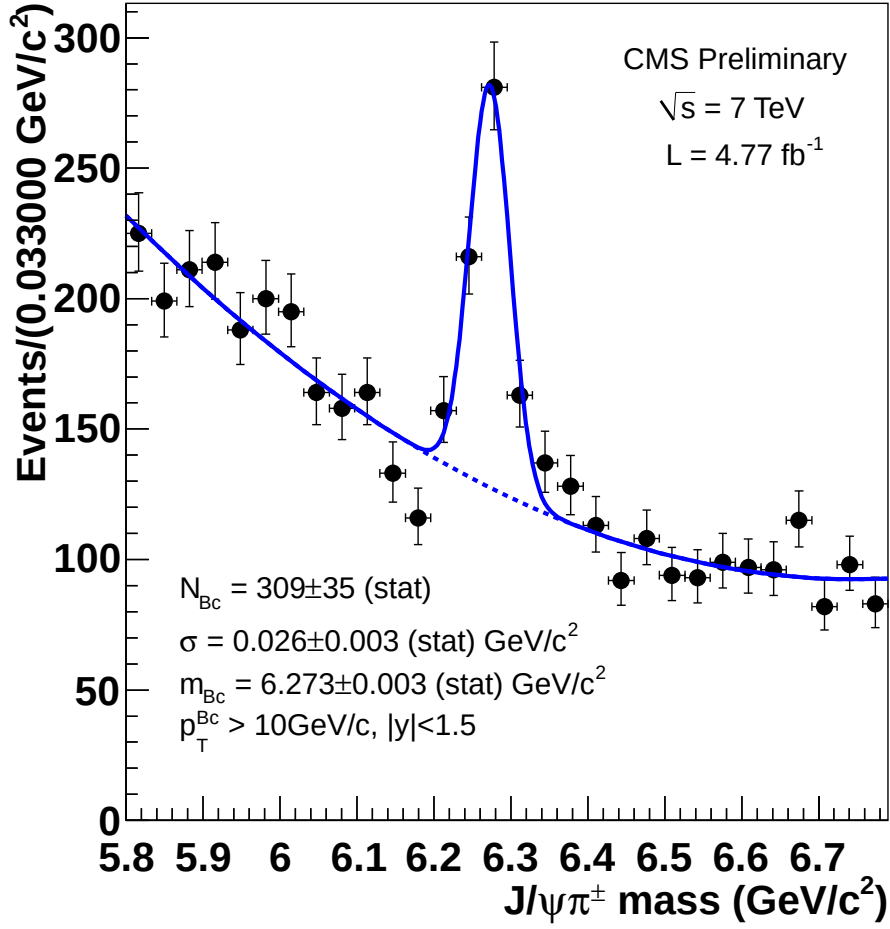


Figure 4: Inv. mass distribution of J/ψ and a pion track for 2011 CMS data. The curve shows the best maximum likelihood fit for the sample. In the fitting, the B_c signal is described by a Gaussian function where we allow the mean and width of Gaussian function to float, while the background is parametrized by 2nd order Chebyshev polynomial function.

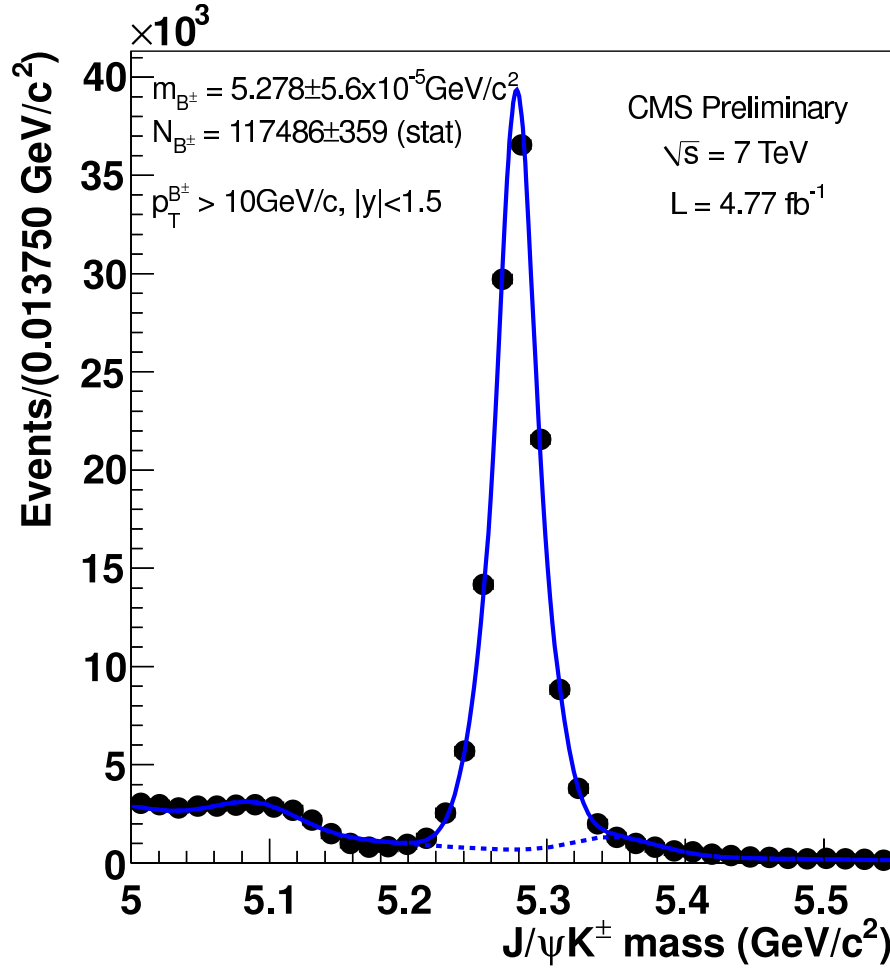


Figure 5: Inv. mass distribution of J/ψ and a kaon track for 2011 CMS data. The curve shows the best maximum likelihood fit for the sample. In the fitting, the $B^\pm$ signal is described by three Gaussian functions, the width of first Gaussian is fixed while all other parameters are floating and background is the combination of exponential function (for flat backgrounds) and a Gaussian which describes the background $B^0 \rightarrow J/\psi K^*(892)$ with missing a pion or kaon track on the left shoulder of $B^\pm$ peak. In the right of $B^\pm$ peak, the contribution of $B^\pm \rightarrow J/\psi \pi^\pm$ process is fixed to the parameters obtained from the MC simulation.

3.3 Data/MC comparison and fit validation

We compared the distributions of $B_c(B^\pm)$ p_T , rapidity, $\cos\alpha$, $L_{xy}/\sigma_{L_{xy}}$, Proper Decay Length (PDL), pions(kaons) p_T and the probability of the kinematic fit between $B_c \rightarrow J/\psi\pi$ ($B^\pm \rightarrow J/\psi K^\pm$) MC signal samples and data. The $B_c(B^\pm)$ candidates in data are after the signal sidebands subtraction, shown in Figure 6 and Figure 7 where sidebands are defined to be the events in two symmetrical bands on the sides of signal peak with the wide of one sigma σ mass resolution. It can be seen that the consistency between the MC samples and data is good.

Validation of the mass fit in Figure 4 is tested by the Pull distribution for toy experiments simulating the final fit. Figure 8 shows the Pull distributions for the B_c signal event yield of the fit, invariant mass and mass resolution. Tests of the Pull distribution illustrate the fitting result of B_c inv. mass spectrum is stable and reliable.

$$Pull(N_{sig}) = \frac{N_{sig}^{fit} - N_{sig}^{true}}{\sigma_N^{fit}} \quad (2)$$

4 Measurements of Cross Sections

The $B_c^+ \rightarrow J/\psi\pi^+$ and $B^+ \rightarrow J/\psi K^+$ production cross sections are measured by the observed B_c^+ (B^+) events corrected by the efficiencies and data luminosity, namely the Equation 3,

$$\frac{d\sigma(pp \rightarrow B_c^+(B^+) X)}{dp_T^{B_c^+(B^+)}} \cdot Br(B_c^+(B^+) \rightarrow J/\psi\pi^+(K^+)) = \frac{N_{sig}}{2 \cdot \varepsilon \cdot Br(J/\psi \rightarrow \mu\mu) \cdot \Delta p_T(|y|) \cdot \mathcal{L}} \quad (3)$$

N_{sig} is the observed B_c ($B^\pm$) events in the given B_c ($B^\pm$) p_T or $|y|$ bin, $Br(J/\psi \rightarrow \mu\mu) = 5.93 \pm 0.06\%$ is obtained from PDG [52], $\Delta p_T(|y|)$ is the bin width. The factor of 2 accounts for our choice of quoting the B_c^+ (B^+) cross sections rather than $B_c^\pm$ and $B^\pm$.

To get the differential cross sections, we separate the selected events into three bins respect to B_c p_T and $|y|$ and nine bins for $B^\pm$ p_T and $|y|$, summarized in Table 5. The bin widths are chosen to maintain statistical uncertainty in each bin to be nearly same.

Table 5: Separation of p_T and $|y|$ bins for differential cross sections $d\sigma B_c^+(B^+)/dp_T$ and $d\sigma B_c^+(B^+)/d|y|$.

B_c	Bin1			Bin2			Bin3		
p_T (GeV/c)	10-16			16-22			> 22		
$ y $	0-0.5			0.5-0.9			0.9-1.5		
$B^\pm$	Bin1	Bin2	Bin3	Bin4	Bin5	Bin6	Bin7	Bin8	Bin9
p_T (GeV/c)	10-14	14-16	16-18	18-20	20-22	22-25	25-29	29-36	> 36
$ y $	0-0.15	0.15-0.3	0.3-0.45	0.45-0.6	0.6-0.75	0.75-0.9	0.9-1.05	1.05-1.25	1.25-1.5

4.1 Signal yields

The signal yields are extracted by the unbinned extended 1D maximum-likelihood fits to the B_c ($B^\pm$) inv. mass distributions in B_c ($B^\pm$) p_T and $|y|$ bins.

In Figure 4 fitting of B_c for single bin, signal is parametrized by a Gaussian function where the Gaussian width and mass mean are allowed to float. In the fitting of B_c for each differential bin, signal is parametrized by a

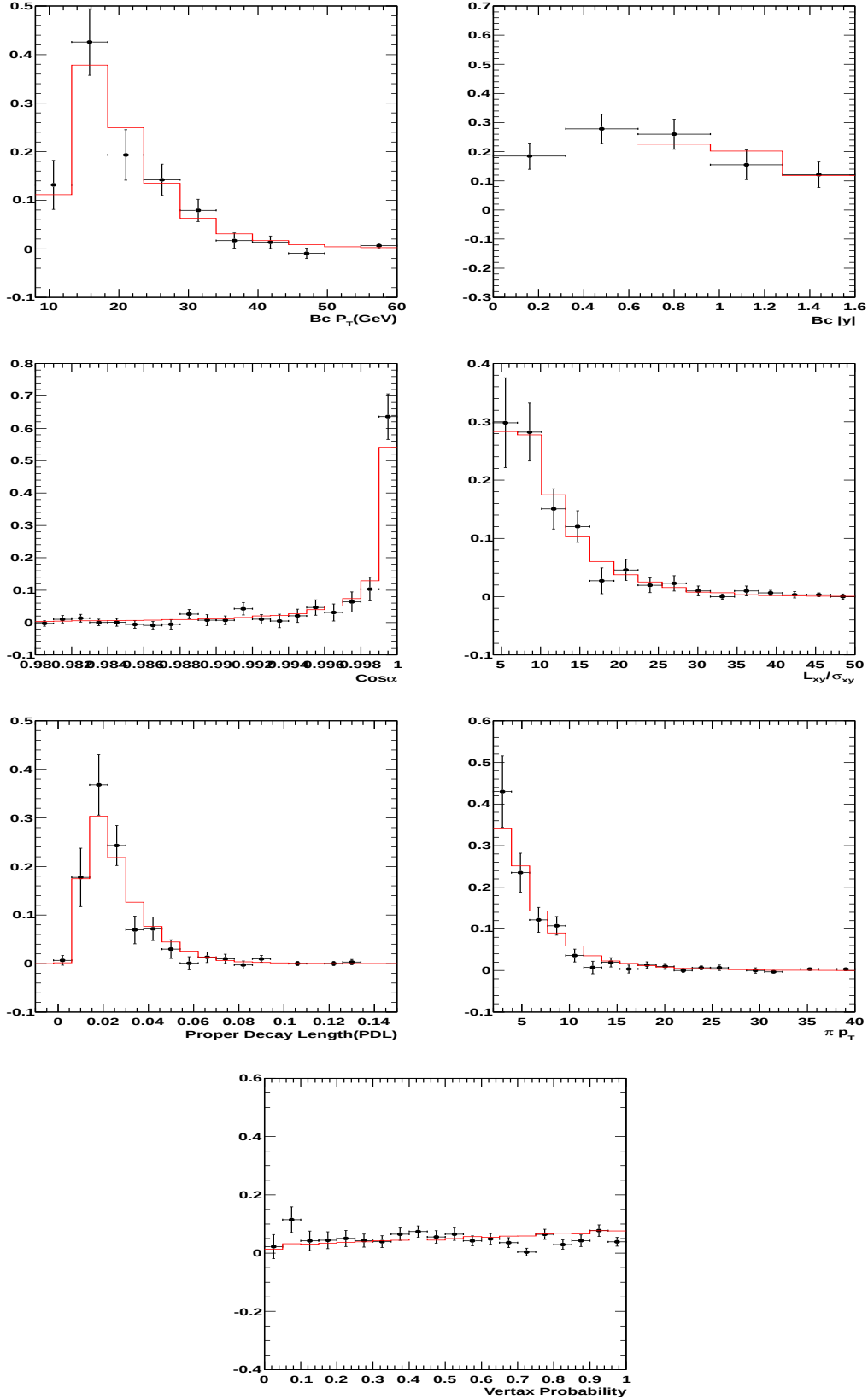


Figure 6: Comparisons between B_c MC signal samples (red histogram) and B_c candidates in data (dots with uncertainty bars) after the signal sidebands subtraction for the distributions of B_c p_T , B_c rapidity, $\cos\alpha$, L_{xy}/σ_{xy} , PDL, $p_T(\pi)$ and the probability of the kinematic fit.

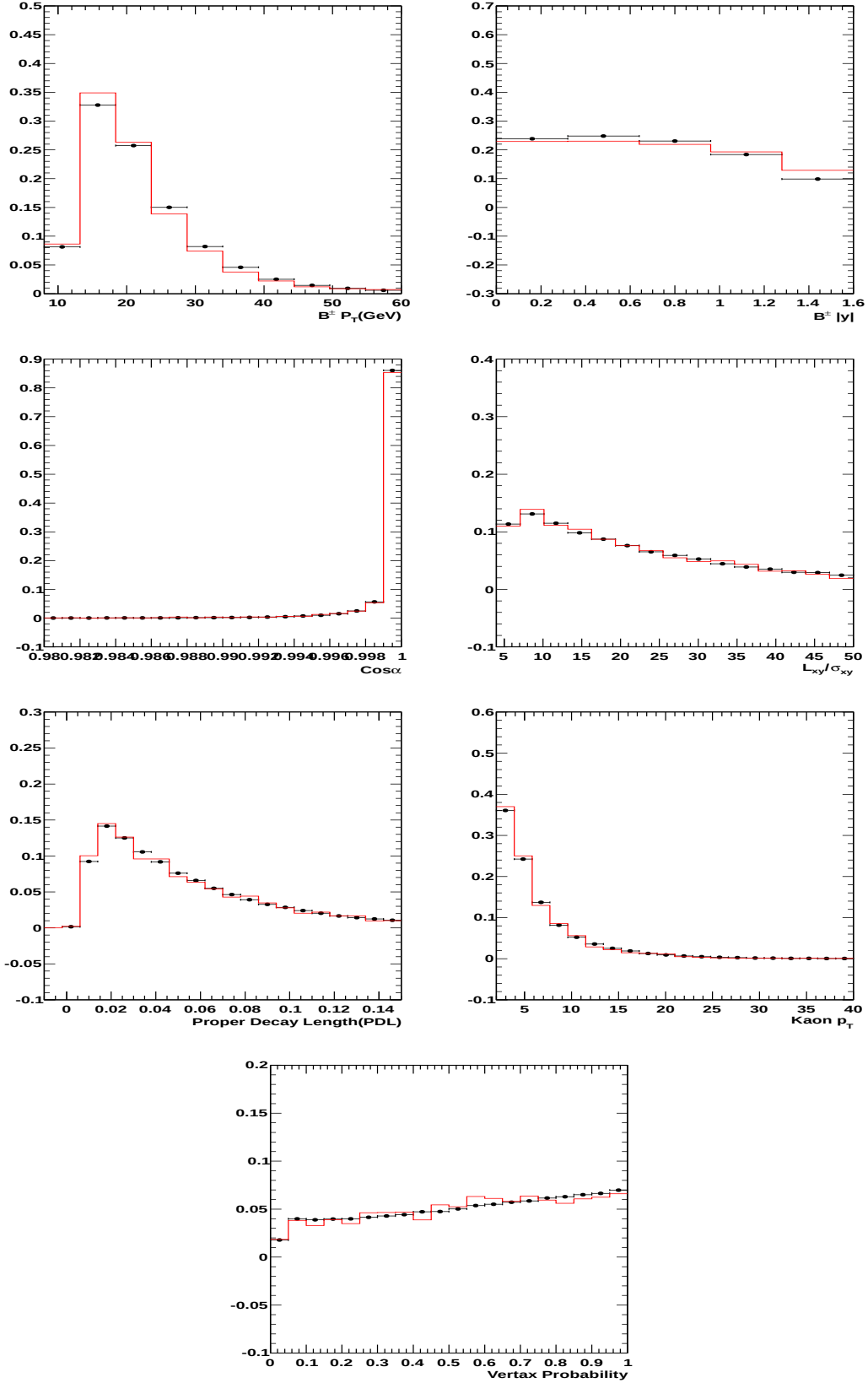


Figure 7: Comparisons between $B^\pm$ MC signal samples (red histogram) and $B^\pm$ candidates in data (dots with uncertainty bars) after the signal sidebands subtraction for the distributions of $B^\pm p_T$, $B^\pm$ rapidity, $\cos\alpha$, $L_{xy}/\sigma_{L_{xy}}$, PDL, $p_T(K)$ and the probability of the kinematic fit.

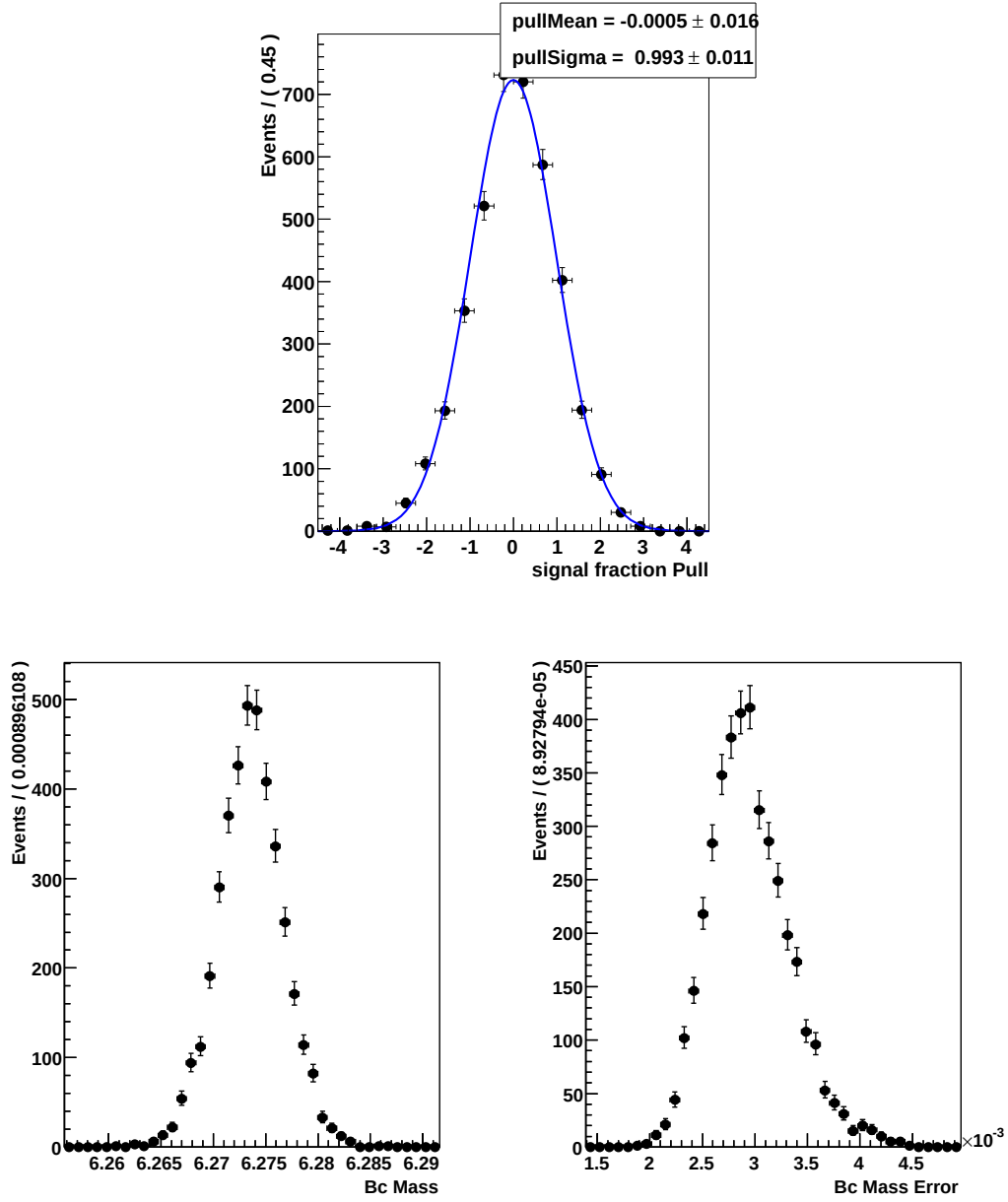


Figure 8: Pull distribution for B_c signal yield (top), invariant mass (bottom left) and mass resolution (bottom right).

Gaussian function where the Gaussian width is allowed to float and the mass mean value is fixed to $6.273 \text{ GeV}/c^2$ which is obtained by the fit in Figure 4 single bin fit of B_c . The background is parametrized by the 2^{nd} order Chebyshev polynomial function. The fitting results and signal yields for B_c p_T bins and $|y|$ bins are shown in Figures 9 and Table 13. For $B^\pm$ single bin, the signal is described by three Gaussian functions where first Gaussian width is fixed while all others parameters are allowed to float. For the $B^\pm$ fit in each differential bin, the signal is described by three Gaussian functions where first Gaussian width is fixed while width of other two Gaussian is allowed to float, mass mean values fixed at $5.27836 \text{ GeV}/c^2$ which is obtained by the fit in Figure 5 single bin fit of $B^\pm$. The background is the combination of an exponential and two Gaussian functions. The enhancement in the left of the $B^\pm$ peak, which is mainly the process $B^0 \rightarrow J/\psi K^*(892)$ with missing a pion or kaon track is described by a Gaussian function with mass mean and width allowed to float. The contribution of $B^\pm \rightarrow J/\psi \pi^\pm$ process in the right of $B^\pm$ peak is described by Gaussian function with fixed parameters obtained from the MC simulation. The fitting results and signal yield for $B^\pm$ p_T bins and $|y|$ bins are shown in Figures 11,12 and Table 14

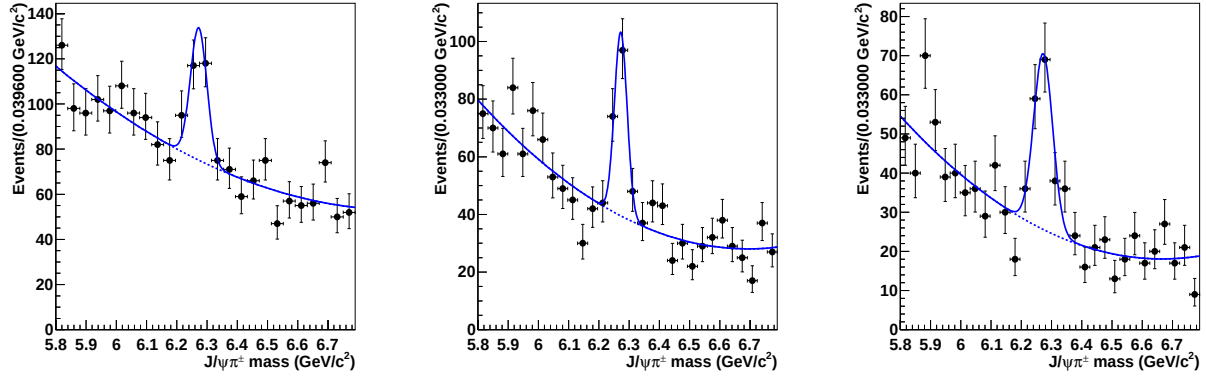


Figure 9: From left to right, fits to the inv. mass distributions of J/ψ and π in data for B_c p_T bins 10-16, 16-22 and $>22 \text{ GeV}/c$.

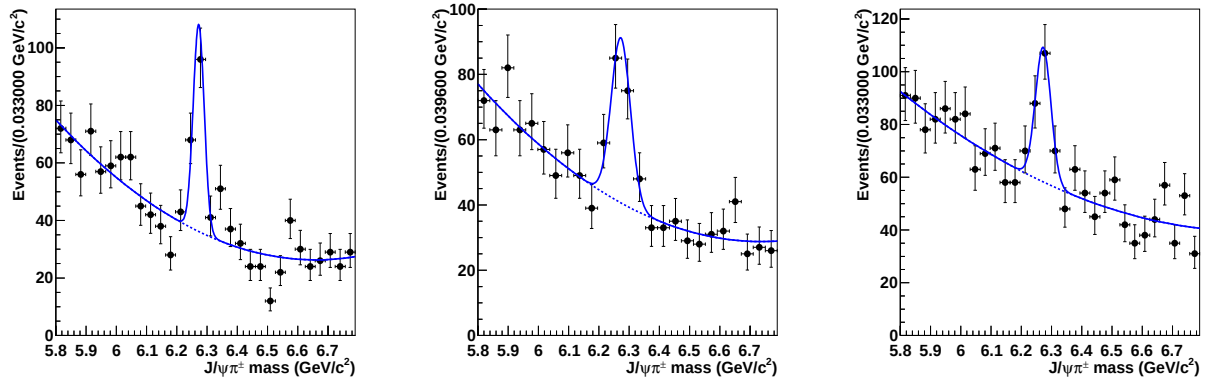


Figure 10: From left to right, fits to the inv. mass distributions of J/ψ and π in data for B_c $|y|$ bins 0.0-0.5, 0.5-0.9 and 0.9-1.5.

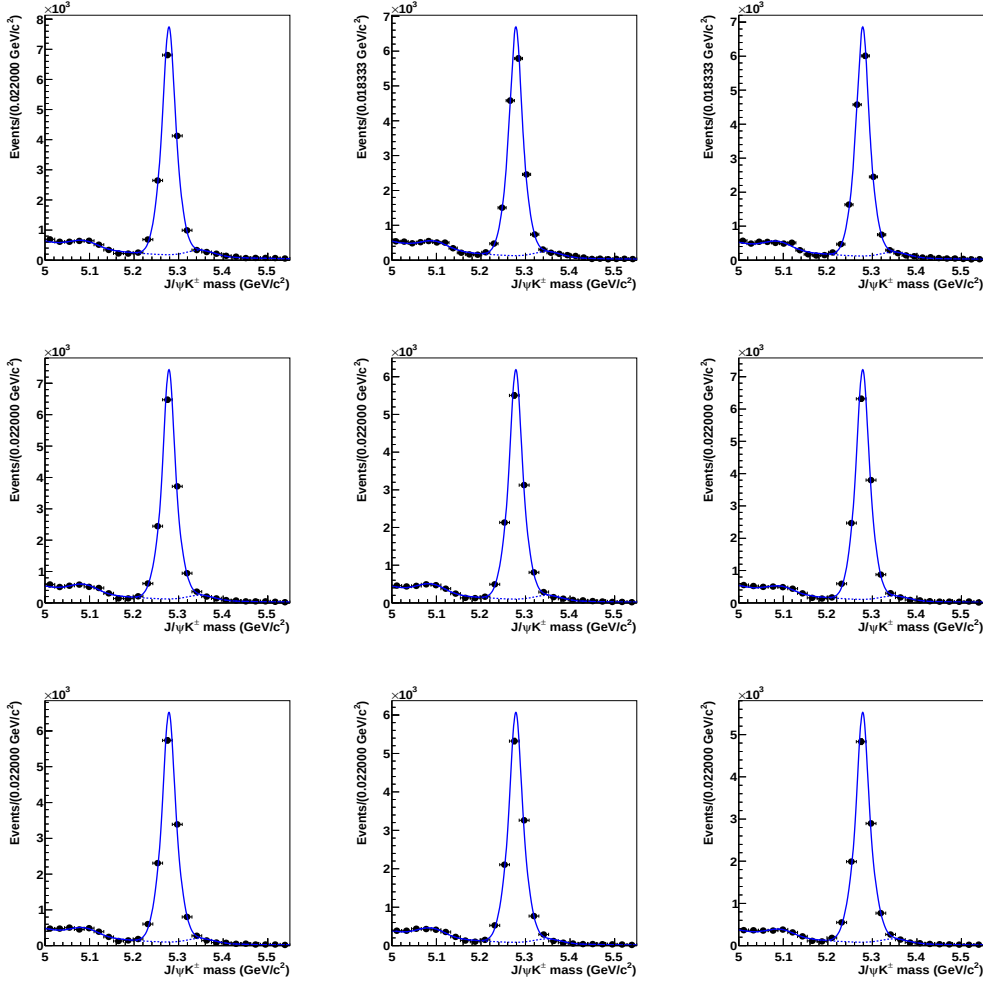


Figure 11: From left to right and top to bottom, fits to the inv. mass distributions of J/ψ and kaon track in data for $B^\pm$ p_T differential bins 10-14, 14-16, 16-18, 18-20, 20-22, 22-25, 25-29, 29-36, > 36 GeV/ c .

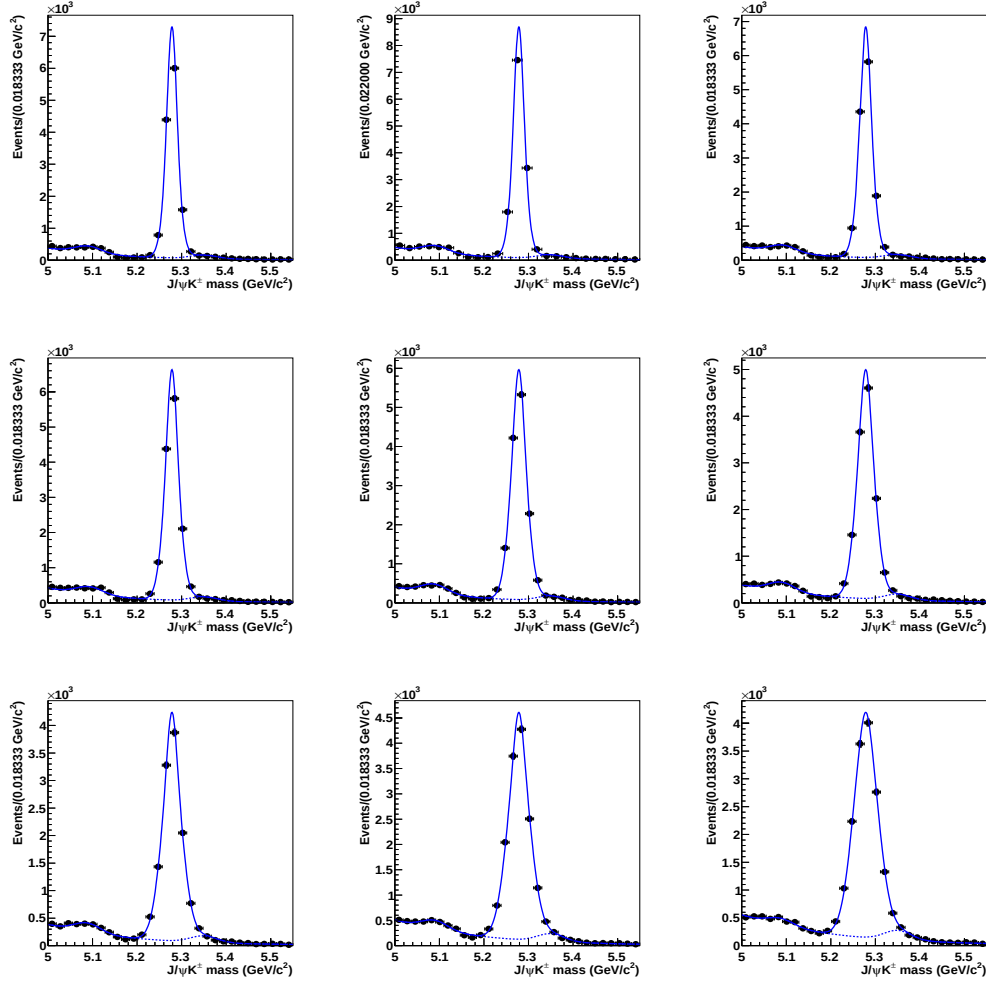


Figure 12: From left to right and top to bottom, fits to the inv. mass distributions of J/ψ and kaon track in data for $B^\pm |y|$ differential bins 0-0.15, 0.15-0.3, 0.3-0.45, 0.45-0.6, 0.6-0.75, 0.75-0.9, 0.9-1.05, 1.05-1.25, 1.25-1.5.

The fit validation for each differential bin is tested by the Pull distribution for 4000 toy experiments simulating the final B_c mass spectrum fit, shown in Figure 13. The Pull distribution is found to be consistent with standard normal distribution which shows stability of our fit .

4.2 Efficiency Determination

The efficiencies for reconstructing $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ and $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ processes are computed from a combination of data-driven technique (T&P) and MC simulation. The efficiency is calculated by Equation 4,

$$\epsilon = Acc. \cdot \epsilon_{MC} \cdot \alpha_{2\mu} \cdot \alpha_{Disp} \quad (4)$$

where $Acc.$ is the detector acceptance which is mainly determined by the dimuon detection capacity of CMS Muon chamber. It is calculated by the signal MC samples without any muon kinematic filter cut. ϵ_{MC} is $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ or $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ reconstruction efficiency with the full cuts for the MC samples except the acceptance. $\alpha_{2\mu}$ is the correction factor for two muon efficiency, and α_{Disp} is the correction factor for the displaced trigger efficiency. In the analysis, two muon trigger identification and tracking efficiencies are calculated by the Tag&Probe method [46] for the data and MC samples, the efficiencies ratio will be taken as the $\alpha_{2\mu}$. The α_{Disp} is the ratio between the displaced J/ψ trigger efficiencies of MC samples and data. The total efficiency is defined as the efficiency to reconstruct a single event $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ or $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$.

4.2.1 Acceptance

The geometrical acceptance of $B_c \rightarrow J/\psi\pi$ ($B^\pm \rightarrow J/\psi K^\pm$), $J/\psi(\mu^+\mu^-)$ process is determined by muon detection capacity of CMS Muon Chamber [45]. It is computed in B_c ($B^\pm$) p_T and $|y|$ bins for the kinematic requirement of $p_T > 10$ GeV/c and $|y| < 1.5$.

The acceptance is calculated by the generator level MC samples which has been described in Section 2. Acceptance is the ratio of the number of events of B_c ($B^\pm$) that satisfies B_c ($B^\pm$) kinematic requirement, $J/\psi p_T$ and $\mu(acc.)$ to the number of events of B_c ($B^\pm$) that satisfies B_c ($B^\pm$) kinematic requirement as in Equation 5

$$Acc. = \frac{N\{B(p_T > 10 \text{ GeV}/c; , |y| < 1.5), J/\psi(p_T > 6.9 \text{ GeV}/c;), \mu(acc.); \}}{N\{B(p_T > 10 \text{ GeV}/c; , |y| < 1.5)\}} \quad (5)$$

We use the the “tracker 50” muon acceptance selections $\mu(acc.)$, which was used in the CMS-Note [46] for the single muon Tag&Probe efficiency calculation. The selections criteria of muon acceptance $\mu(acc.)$ are listed in Section 3.1. Figure 14 shows the distribution of two-dimensional acceptance for B_c p_T and $|y|$. Acceptance in B_c p_T and $|y|$ bins are shown in Figure 15. Figure 16 shows the distribution of two-dimensional acceptance for $B^\pm$ p_T and y . Acceptance in $B^\pm$ p_T and $|y|$ bins are shown in Figure 17.

4.2.2 Monte Carlo efficiency

The ϵ_{MC} , as expressed in Eq. 4, is full $B_c^+ \rightarrow J/\psi(\mu^+\mu^-)\pi^+$ or $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ reconstruction efficiency determined by the MC samples except acceptance. It is ratio between the signal MC events after the full cuts, ΔR match with generated and the accepted MC B_c ($B^\pm$) events.

$$\epsilon_{MC} = \frac{N(Acc., Disp., \text{all cuts}, \Delta R \text{ match})}{N(Gen., Acc.)} \quad (6)$$

Figure 18 shows the MC $B_c^+ \rightarrow J/\psi(\mu^+\mu^-)\pi^+$ reconstruction efficiency, ϵ_{MC} , for different B_c p_T and $|y|$ bins. Figure 19 shows the MC $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ reconstruction efficiency, ϵ_{MC} , for different $B^\pm$ p_T and $|y|$ bins.

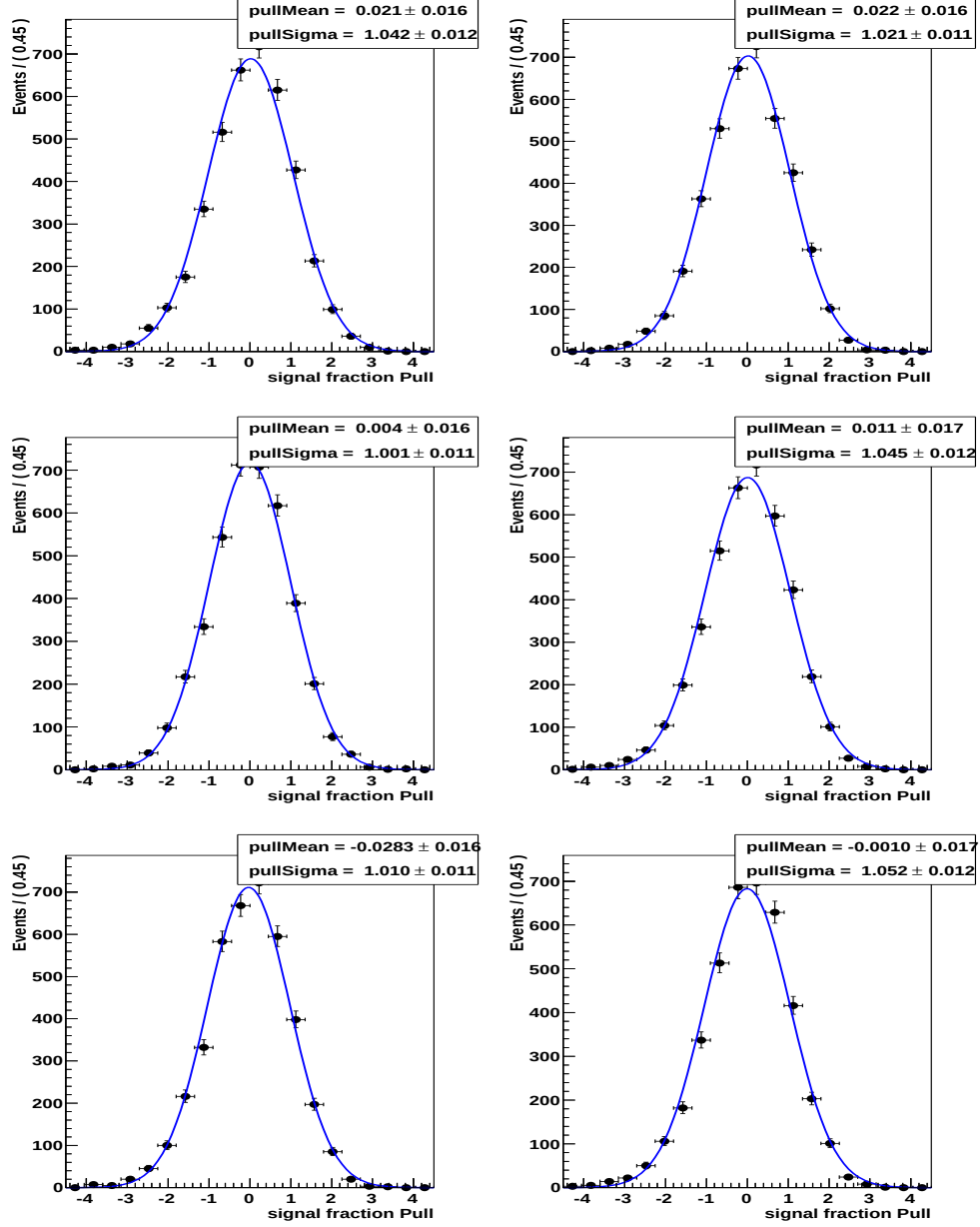


Figure 13: Pull distributions $(n_{fit} - \langle n \rangle) / \sigma$ for 4000 toy experiments simulating the final fit. From top to bottom, the left are for $B_c p_T$ differential bins 10-16, 16-22 and > 22 GeV/c, the right are for $B_c |y|$ differential bins 0-0.5, 0.5-0.9 and 0.9-1.5.

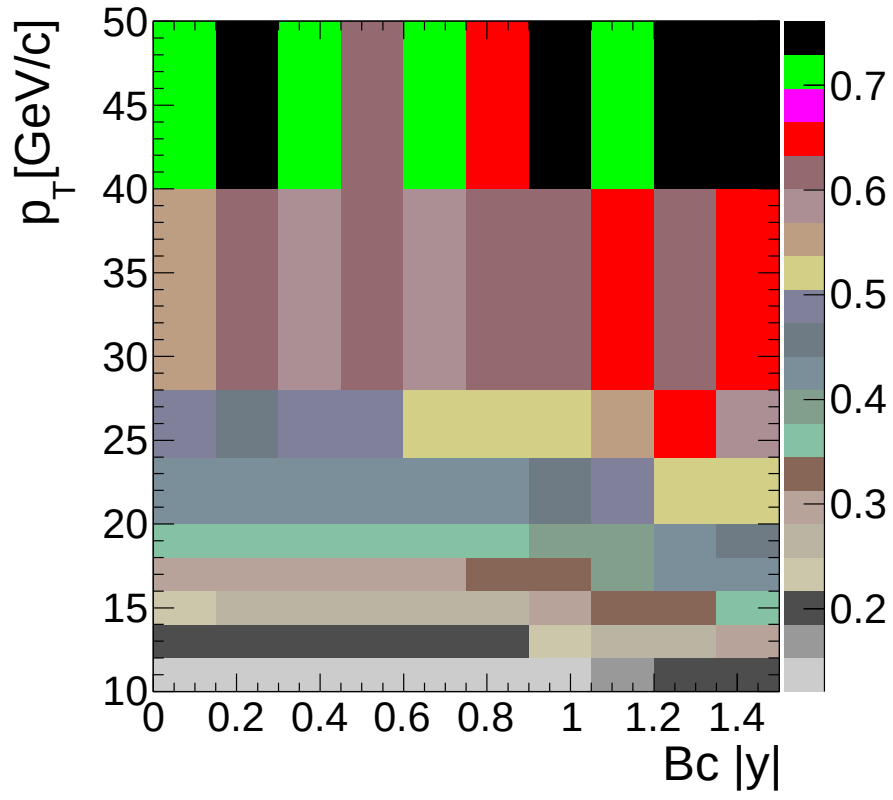


Figure 14: The acceptance of $B_c \rightarrow J/\psi\pi$, $J/\psi \rightarrow \mu^+\mu^-$ process in $B_c p_T$ and $|y|$ dimensions.

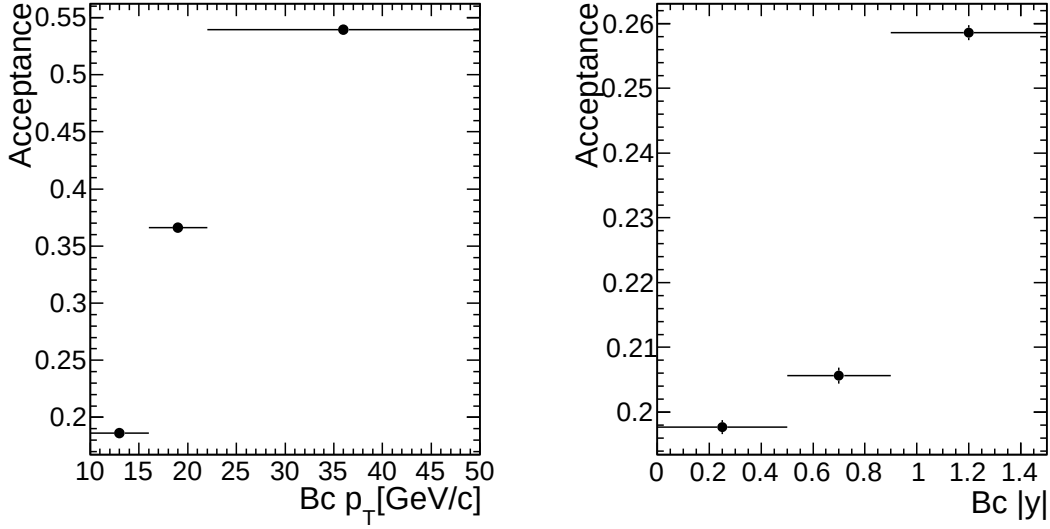


Figure 15: B_c acceptance as a function of $B_c p_T$ (left) and rapidity (right), where the uncertainties reflect the MC statistics.

4.2.3 Muon efficiency and correction

Efficiency of dimuon selection $\varepsilon_{2\mu}$, is measured by using the “Tag and Probe” technique [46] for the MC samples and data. The expressions used to compute $\varepsilon_{2\mu}$ are:

$$\varepsilon_{\mu} = \varepsilon_{tracking} \times \varepsilon_{ID} \times \varepsilon_{Qual} \times \varepsilon_{L1.L2} \times \varepsilon_{L3} \quad (7)$$

$$\varepsilon_{D\mu} = \varepsilon_{\mu_1} \times \varepsilon_{\mu_2} \quad (8)$$

where μ_1 and μ_2 refer to the two muons decaying from $B_c(B^{\pm}) \rightarrow J/\psi\pi(K^{\pm})$, $J/\psi \rightarrow \mu^+\mu^-$. $\varepsilon_{tracking}$ is the muon track reconstruction efficiency, ε_{ID} is the muon identification efficiency, ε_{Qual} is the efficiency of the muon quality cuts, $\varepsilon_{L1.L2}$ and ε_{L3} are the muon trigger efficiencies of the level 1, level 2 and level 3 triggers. The product of the single muon efficiencies gives the dimuon efficiency $\varepsilon_{D\mu}$.

The “Tag and Probe” study has been performed earlier with same data set using the package `MuonAnalysis/TagAndProbe`. In this analysis, same muon selection criteria is used as in the reference [46]. Muon kinematics requirement is the “tracker 50” muon acceptance cut. For the $\varepsilon_{tracking}$, we follow the strategy outlined in reference [46] and apply an efficiency of $99.0 \pm 1.0\%$ as recommended by muon POG. The Tag&Probe data files (listed in the link [51]) of the single muon efficiency we used for ε_{ID} and ε_{Qual} calculations are the follows,

- `MuonID_pt_abseta_runA_inclMay10-TrkCuts_15Nov2011_corrected.root`
- `MuonQual_pt_abseta_runA_inclMay10-TrkCuts_15Nov2011_corrected.root`

We split the calculation of the trigger efficiency into two parts. The first part involves the triggers which are based only on single muon properties and second part includes those which rely on dimuon variables (Displaced trigger). The way of calculating the efficiency of the dimuon trigger along with the requirements (confidence level of the dimuon vertex fit, dimuon flight length significance and dimuon pointing angle) has been elaborated in Section 4.2.4. The factorisable trigger efficiency portion of single muon $\varepsilon_{L1.L2}$ and ε_{L3} , are evaluated with Tag-Probe technique as described in the studies [46].

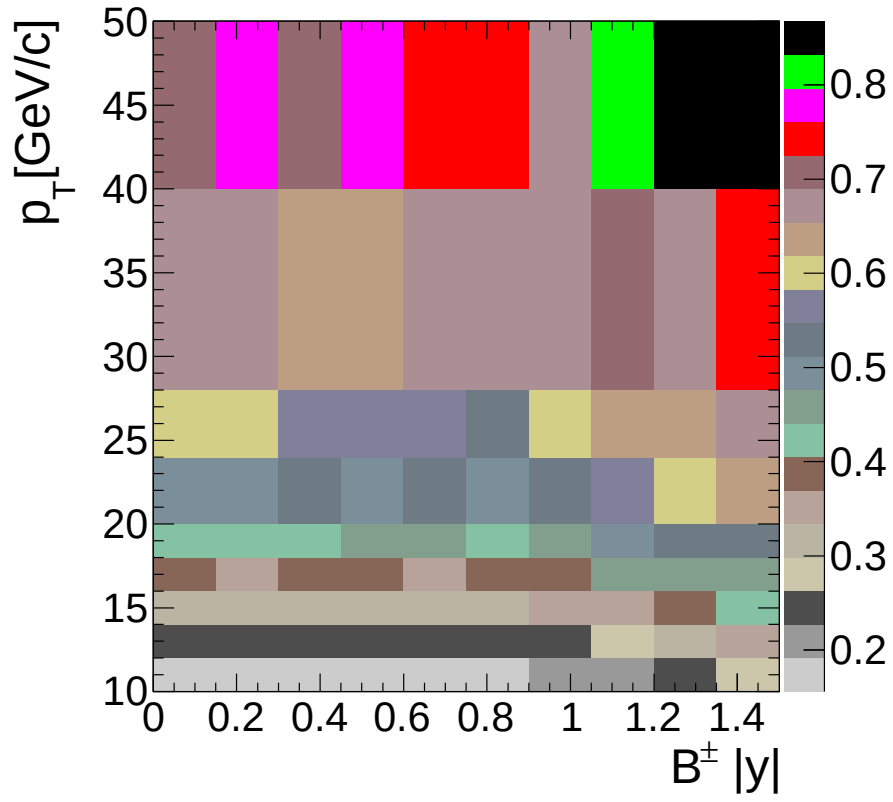


Figure 16: The acceptance of $B^\pm \rightarrow J/\psi K^\pm$, $J/\psi \rightarrow \mu^+ \mu^-$ process in $B^\pm p_T$ and $|y|$ dimensions.

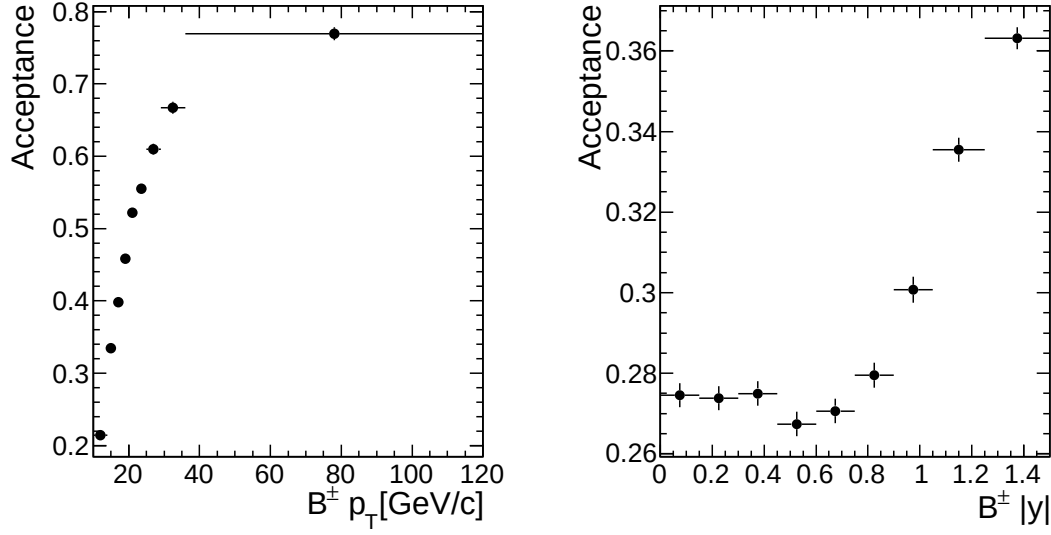


Figure 17: $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ acceptance as a function of $B^\pm p_T$ (left) and absolute rapidity $|y|$ (right), where the uncertainties reflect the MC statistical uncertainties.

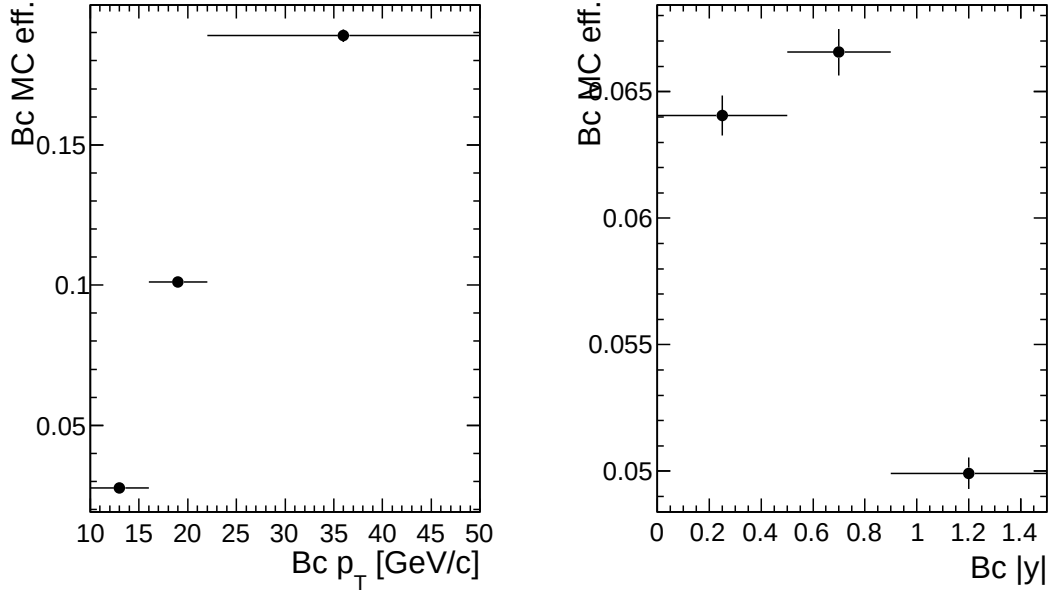


Figure 18: The MC efficiency ϵ_{MC} for different $B_c p_T$ (left) and $|y|$ (right) bins.

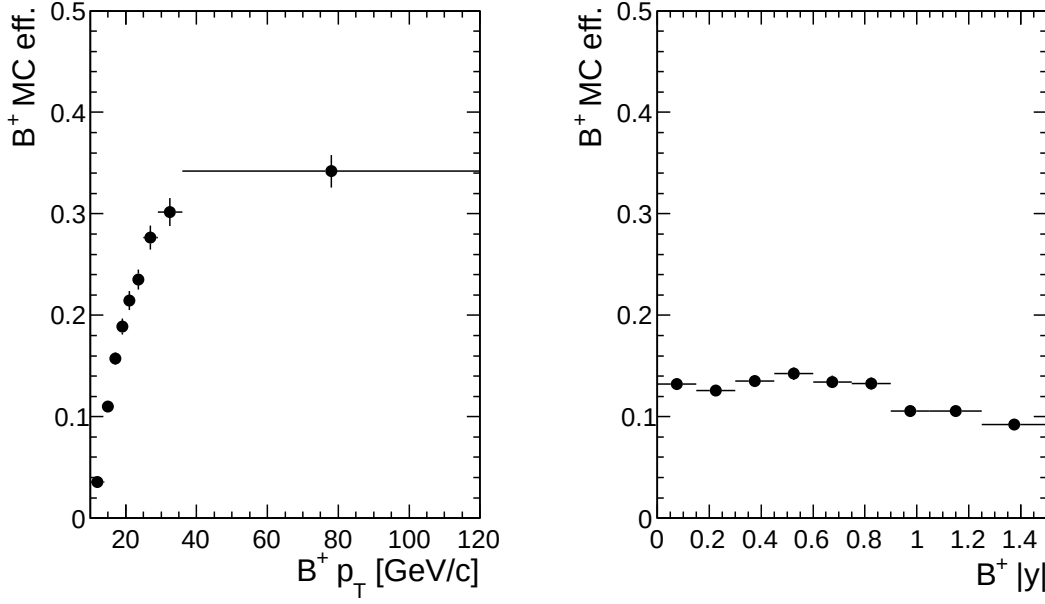


Figure 19: The MC efficiency ϵ_{MC} for different $B^\pm p_T$ (left) and $|y|$ (right) bins.

Tag and Probe studies [46] have shown that there is no obvious change in $\epsilon_{L1.L2}$ with different $L1$ trigger seed `L1_DoubleMu0` (run period 165088-173236) and `L1_DoubleMu0_HighQ` (from the 3E33 trigger menu -run 173236). So the changes regarding to the $L1$ seed in the efficiency calculation can be ignored.

While performing the calculation of the muon trigger efficiency, the issue of the correlation of the muon pair can not be ignored. The correlation has been tagged as an effect of muon pair that crosses in the two muon system. To reject fake muon, an algorithm is applied in the muon reconstruction to remove the muon candidates with shared hits. Shared hits from true muon candidates can happen for the muon pairs that cross each other in the muon system. As such, the “cowboy” muon pairs that bend in the magnetic field toward each other ($\phi_{\mu^+} - \phi_{\mu^-} > 0$) have been shown to have a lower efficiency than the “sea-gull” muon pairs that bend away from each other ($\phi_{\mu^+} - \phi_{\mu^-} < 0$). To overcome this effect, we calculated the muon $L1.L2$ and $L3$ efficiencies for cowboy and seagull muons separately. The $L1.L2.L3$ triggers efficiencies for cowboy-only and seagull-only muons are shown in Figure 20.

In computing the efficiency for each $B_c p_T$ and y bin, Tag and Probe results of seagull (cowboy) muons are used for true seagull (cowboy) muon pairs in the signal MC events, depending on whether the event is a seagull or cowboy. The data files of the single muon efficiency we used are the follows [51]:

- `L1L2Dimuon0Jpsi_pt_abseta_cowboys_TrkCuts_6Nov2011_corrected.root`
- `L1L2Dimuon0Jpsi_pt_abseta_seagulls_run1_TrkCuts_12Nov2011_corrected.root`
- `L3Dimuon0Jpsi_pt_abseta_cowboys_TrkCuts_6Nov2011_corrected.root`
- `L3Dimuon0Jpsi_pt_abseta_seagulls_run1_TrkCuts_12Nov2011_corrected.root`
- `L1L2_pt_abseta_MC_cowboys_TrkCuts_200oct2011_corrected.root`
- `L3_pt_abseta_MC_cowboys_TrkCuts_200oct2011_corrected.root`

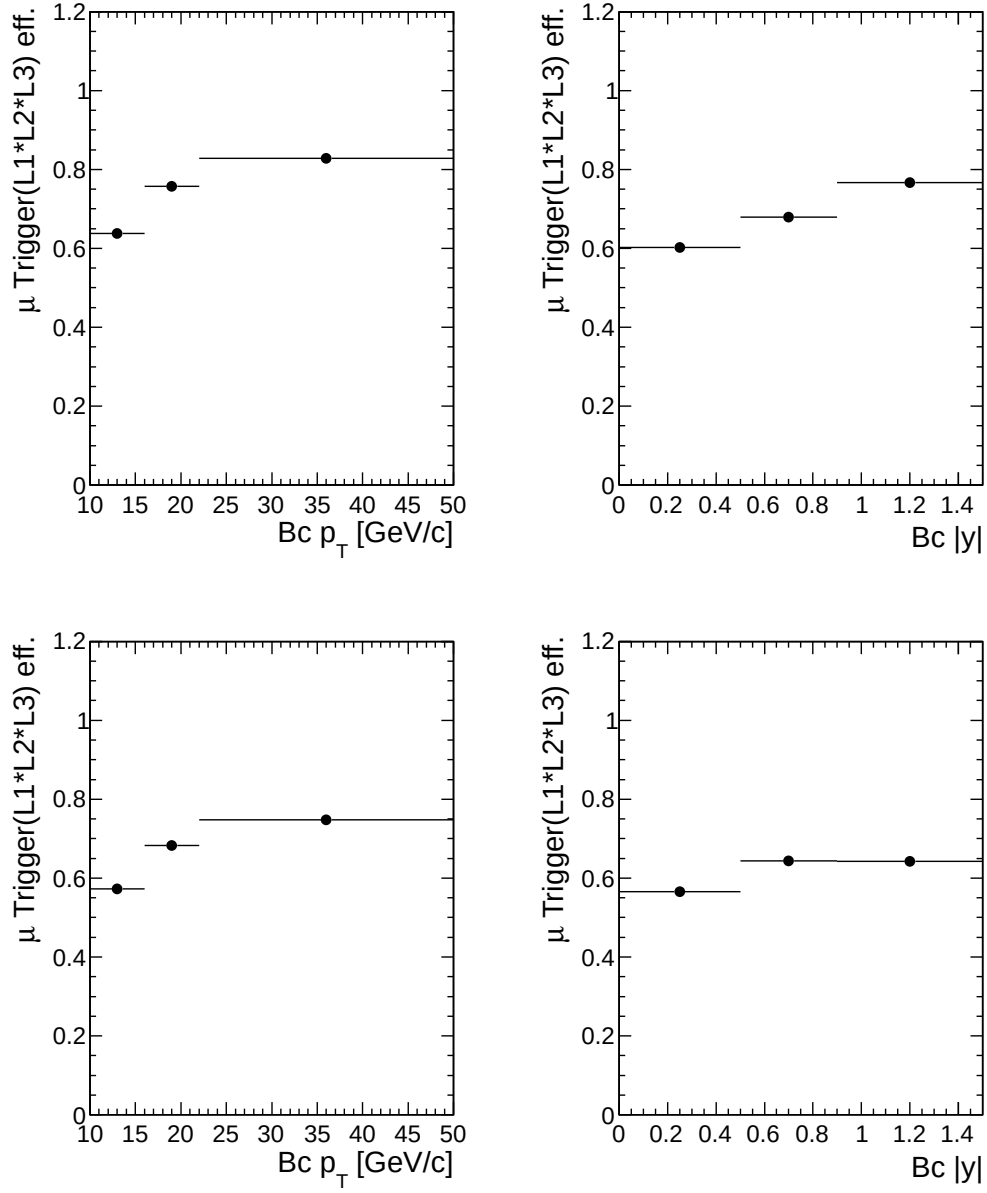


Figure 20: $L1.L2.L3$ trigger efficiencies as a function of $B_c p_T$ (left) and $|y|$ (right) for seagulls (top) and cowboys (bottom) muons. In each case, dimuon efficiency is computed from the product of the two single muon efficiencies.

Figure 21 shows the $L1 \cdot L2$ and $L3$ trigger efficiencies in $B_c p_T$ and $|y|$ bins.

To obtain total two muons efficiency $\varepsilon_{2\mu}$ for each $B_c (B^\pm) p_T$ and $|y|$ bin, we use the approach being expressed in Equation 9.

$$\varepsilon_{2\mu}(p_T, |y|) = N(p_T, |y|) / \sum_{i=0}^{i=N(p_T, |y|)} \frac{1}{\varepsilon_{D\mu, i}} \quad (9)$$

Where $N(p_T, |y|)$ is the number of signal events with two muons within acceptance in $B_c (B^\pm) p_T$ or $|y|$ bins and $\varepsilon_{D\mu, i}$ is dimuon efficiency in each event described in Equation 8. Figure 22 shows the muon efficiencies of each component in Equation 7 for the data, while Figure 23 shows the muon efficiencies of each component in Equation 7 for the MC samples. Dimuon Tag&Probe efficiencies for the data and MC samples are shown in Figure 24. The correction factors of muon Tag&Probe efficiency $\alpha_{2\mu} = \frac{\varepsilon_{2\mu-Data}}{\varepsilon_{2\mu-MC}}$ for different $B_c p_T$ and $|y|$ bins are shown in Figure 25.

Figure 26 shows the $B^\pm$ muon Tag&Probe efficiencies for the data and MC samples, Figure 27 shows the correction factors of muon Tag&Probe efficiency $\alpha_{2\mu} = \frac{\varepsilon_{2\mu-Data}}{\varepsilon_{2\mu-MC}}$ for different $B^\pm p_T$ and $|y|$ bins.

4.2.4 Displaced trigger efficiency and correction

The efficiency due to the displaced portion of the trigger is evaluated by selecting events in data (prescaled) that pass a prompt dimuon trigger (HLT_Dimuon0_Jpsi_NoVertexing_v2). Since displaced trigger is based entirely on the dimuon selections, we use an inclusive dimuon data sample to evaluate the displaced trigger efficiency to increase the available statistics. We apply the offline cuts relevant to the $J/\psi \rightarrow \mu^+\mu^-$, it is an assumption that the $B_c (B^\pm)$ reconstruction requirements do not effect displaced trigger efficiency. Since the offline cuts are always at least as tight as the on-line trigger cuts, this efficiency is expected to be high.

The efficiency is calculated as the fraction of $J/\psi \rightarrow \mu^+\mu^-$ events passing the displaced trigger cuts and prompt trigger to the events passing the prompt trigger. It is described as Equation 10,

$$\varepsilon_{Displaced} = \frac{N(J/\psi \text{ cuts}, \text{Prompt Trigger}, \text{Displaced Trigger})}{N(J/\psi \text{ cuts}, \text{Prompt Trigger})} \quad (10)$$

Table 6 shows the displaced trigger efficiencies in $J/\psi p_T$ and $|\eta|$ bins for different run periods in 2011. The last column shows the average efficiency weighted by the integrated luminosity of each run period. The average J/ψ displaced trigger efficiency then calculated to its mother $B_c (B^\pm)$ in the calculation of $\varepsilon_{B_c} (\varepsilon_{B^\pm})$.

Table 7 shows the displaced trigger efficiency in $J/\psi p_T$ and $|\eta|$ bins for the B_c MC samples. The MC displaced trigger efficiencies for $B^\pm$ are listed in Table 8 for different p_T and $|y|$ bins. Figure 28 shows the correction factors of displaced trigger efficiency $\alpha_{Displaced} = \frac{\varepsilon_{Displaced-Data}}{\varepsilon_{Displaced-MC}}$ for different $B_c (B^\pm) p_T$ and $|y|$ bins.

4.2.5 Total efficiency

We summarize the efficiency measurement of $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ process, being defined by Equation 4, for each $B_c p_T$ and $|y|$ bin in Table 9. Figure 29 shows the total efficiency as a function of $B_c p_T$ and absolute rapidity $|y|$. Following the same steps for $B^\pm \rightarrow J/\psi(\mu^+\mu^-)K^\pm$ we calculated the total efficiency. The Figure 30 and Table 10 are referred for efficiency results regarding to $B^\pm$. The uncertainties are statistical uncertainties of the Tag&Probe sample and $B_c^+ \rightarrow J/\psi(\mu^+\mu^-)\pi^+$, $B^+ \rightarrow J/\psi(\mu^+\mu^-)K^+$ MC samples.

4.3 Systematic Uncertainties

Main sources of systematic uncertainties on the cross section measurements are due to uncertainties on the number of observed signal events, reconstruction efficiencies and integrated luminosity. The systematic uncer-

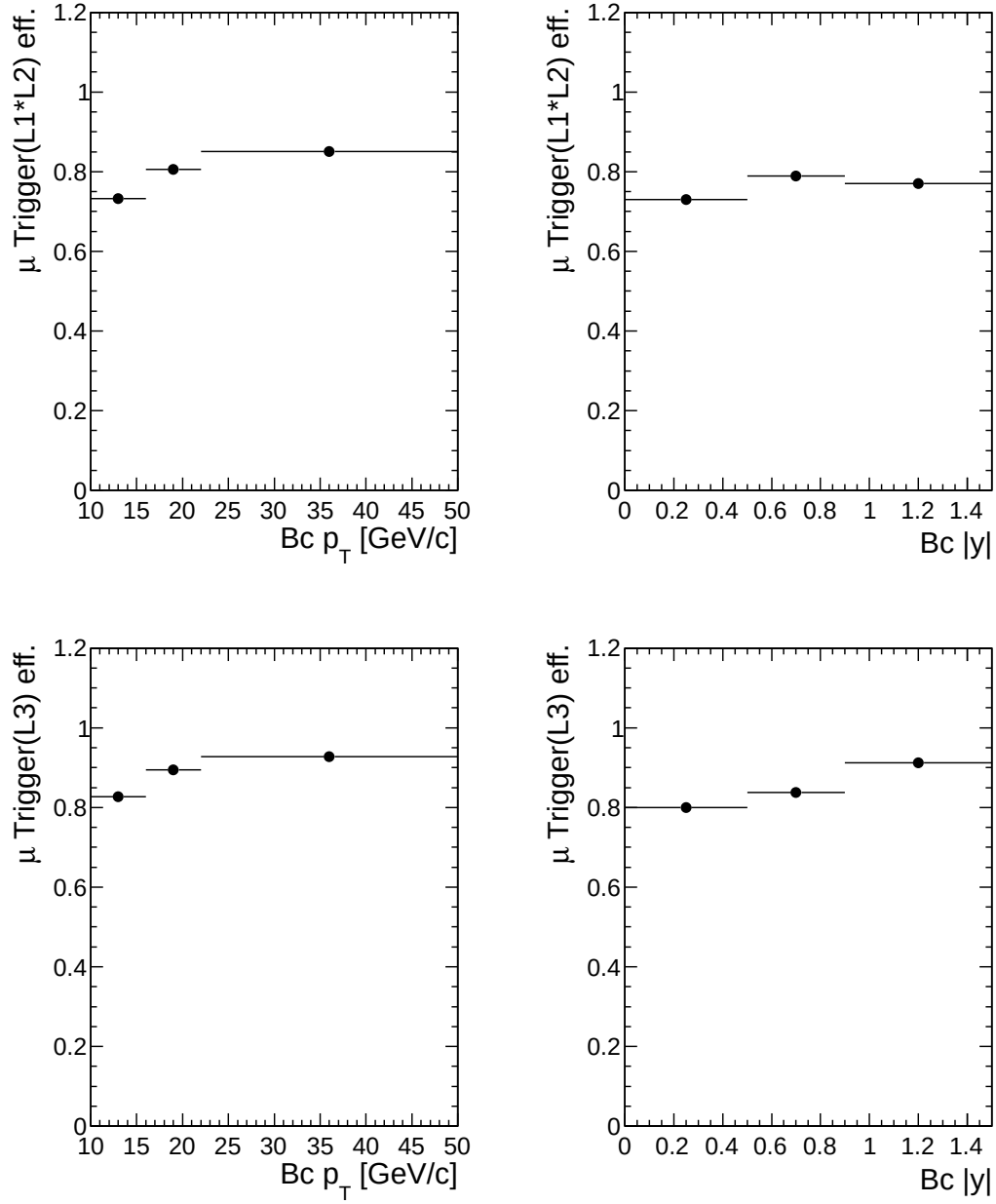


Figure 21: Efficiencies of trigger $L1 \cdot L2$ (top) and $L3$ (bottom) as a function of $B_c p_T$ (left) and y (right). Dimuon efficiency is computed from the product of the two single muons efficiencies.

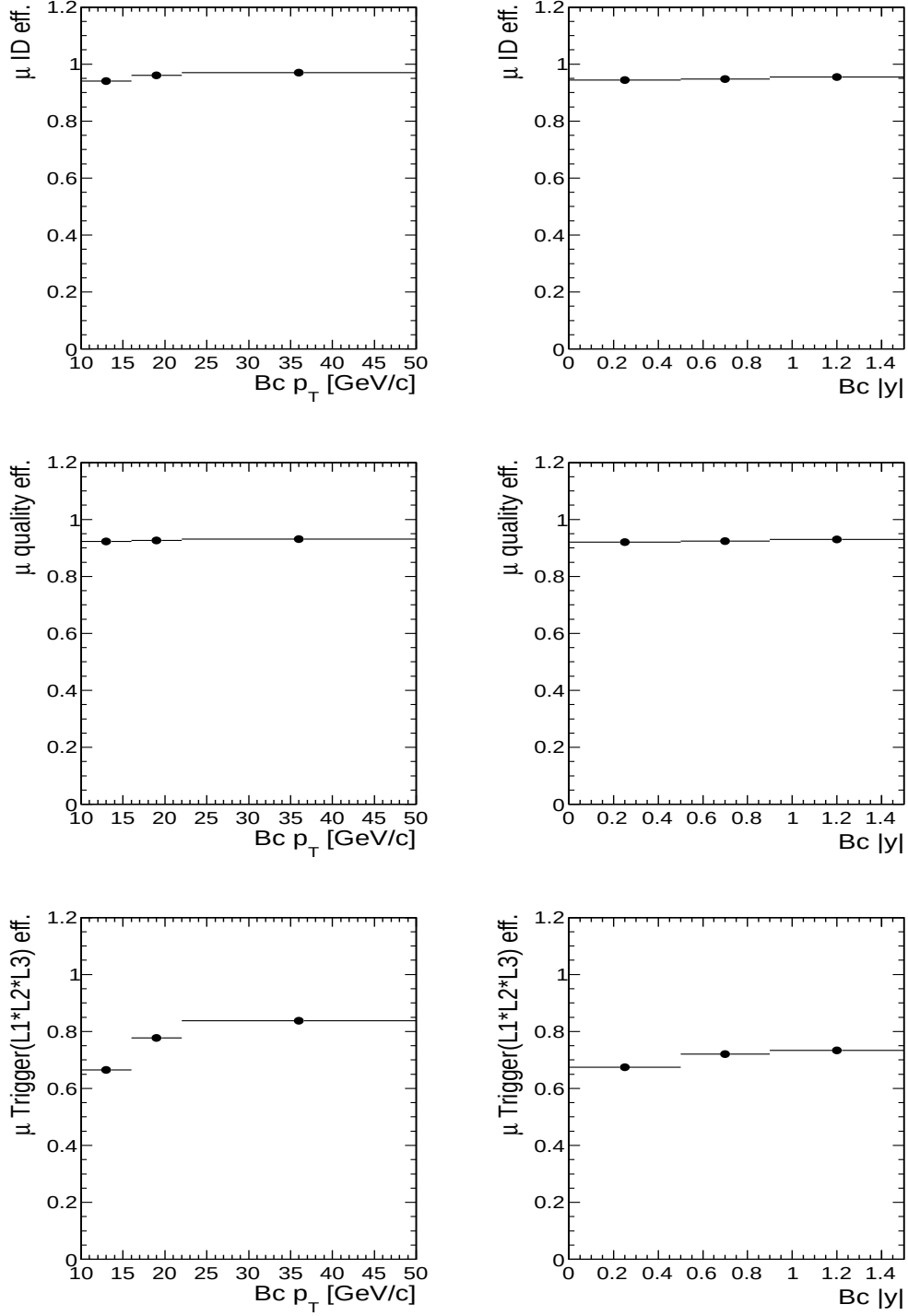


Figure 22: Efficiencies as a function of $B_c p_T$ (left) and $|y|$ (right) for muon identification (top), muon quality (middle) and trigger ($L1 \cdot L2 \cdot L3$) (bottom) for the B_c MC samples.

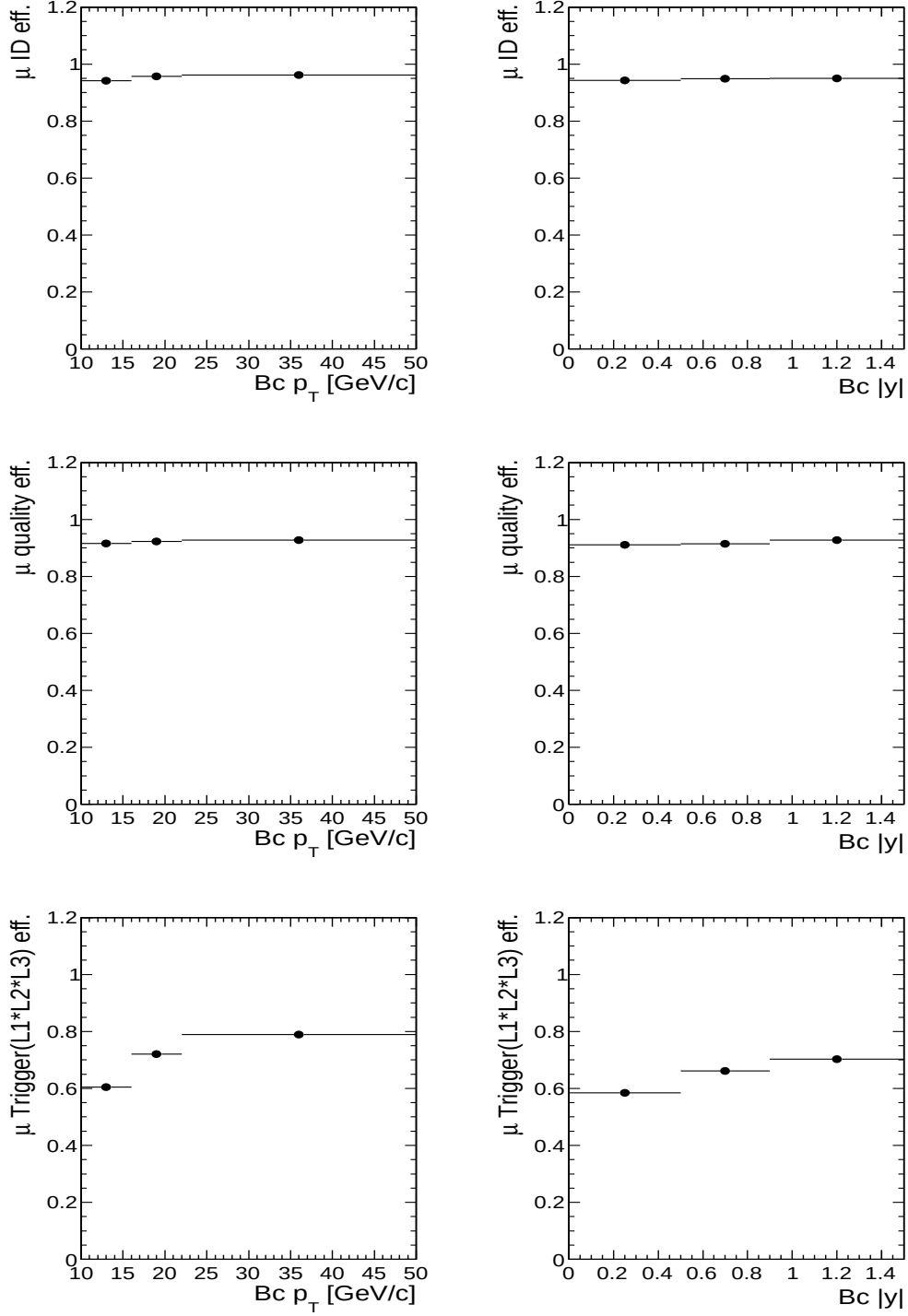


Figure 23: Efficiencies as a function of $B_c p_T$ (left) and $|y|$ (right) for muon identification (top), muon quality cuts (middle) and the Trigger ($L1 \cdot L2 \cdot L3$) (bottom) for the data.

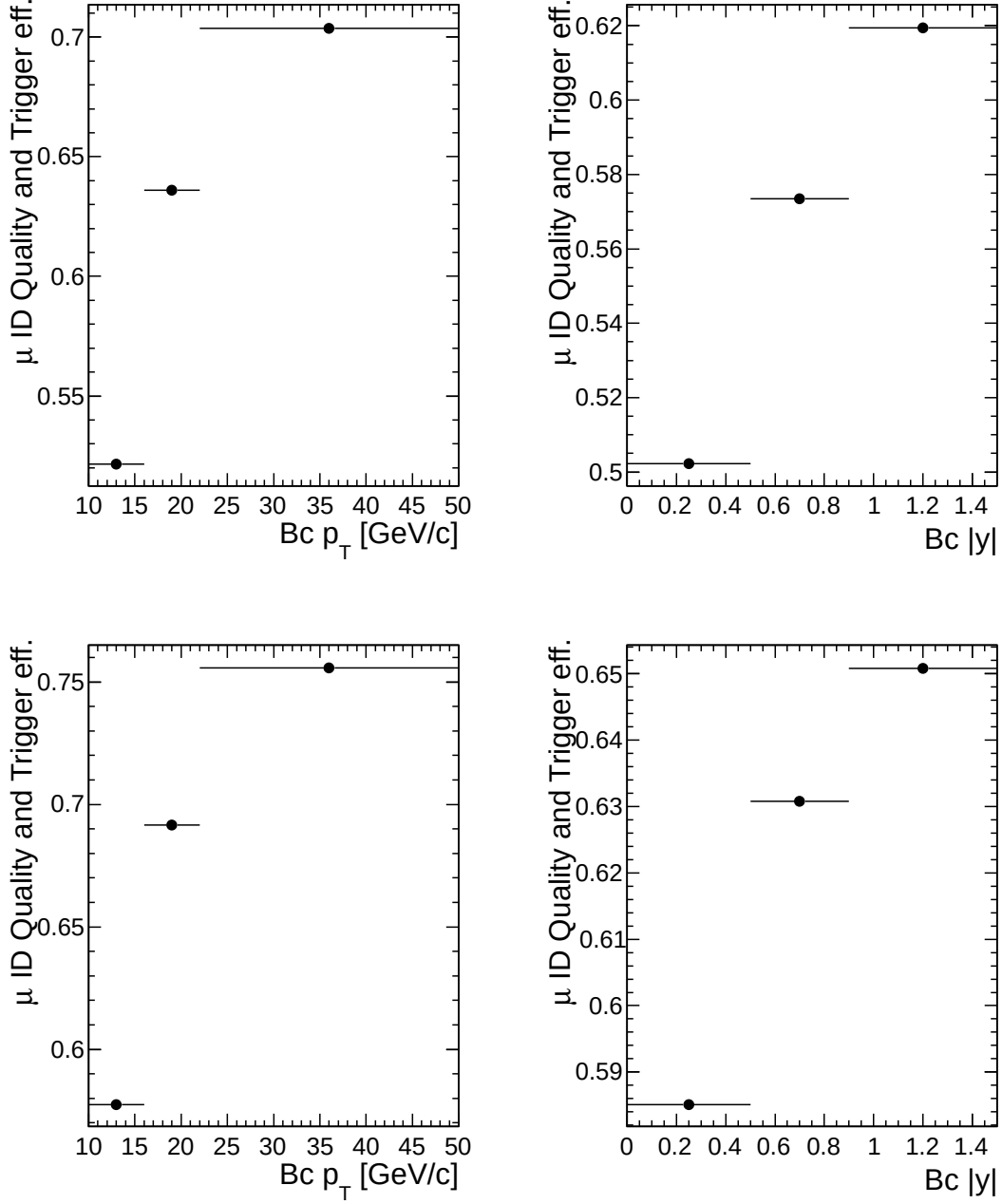


Figure 24: Dimuon efficiencies as a function of $B_c p_T$ (left) and $|y|$ (right). Upper plots are for the data and lower plots are for the MC samples.

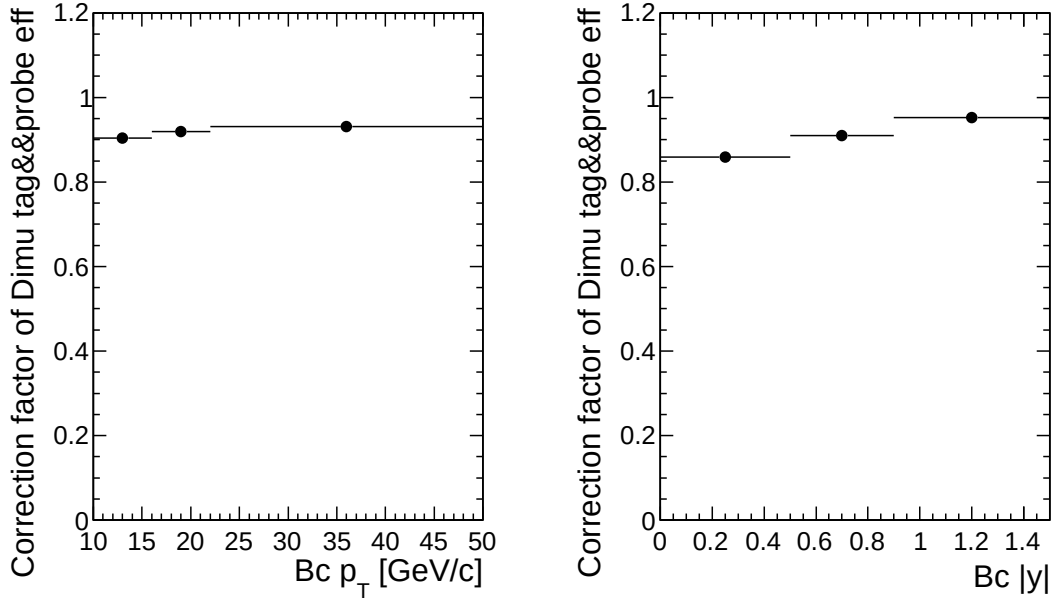


Figure 25: The correction factors of muon Tag&Probe efficiency $\alpha_{2\mu}$ for different $B_c p_T$ and $|y|$ bins.

tainties can also be different in different $B_c(B^\pm) p_T$ and $|y|$ bins. In order to incorporate this effect, we have calculated systematics bin by bin. The systematic uncertainties for the cross section measurements includes the following uncertainties.

4.3.1 Uncertainties on signal shape and yields

The uncertainty affecting the shape of signal arise from non perfect knowledge of signal PDF and resolution. In the B_c mass spectrum fitting, we use single Gaussian function to describe signal, the resolution float. In order to estimate uncertainty on signal shape and yield, we fit B_c mass spectrum with double Gaussian function and allowed resolution to float. The difference in signal yield in each p_T and $|y|$ bin is quoted as signal shape and yields uncertainty.

In B^+ mass spectrum fitting we use three Gaussian functions to describe the signal, the resolution float. In order to estimate uncertainty on signal shape and yield, we fit B^+ mass spectrum with double Gaussian function and allowed resolution to float. The difference in signal yield in each p_T and $|y|$ bin is quoted as signal shape and yields uncertainty.

4.3.2 Uncertainties on background shapes and yields

In the B_c mass spectrum fitting, the background is described to be 2^{nd} order Chebyshev polynomial function. In order to estimate uncertainty on background shapes, we replaced background PDF 2^{nd} order Chebyshev polynomial function with exponential function keeping same signal PDF. The difference in signal yield in each p_T and $|y|$ bin is quoted as background shapes and yields uncertainty.

In the B^+ mass spectrum fitting, the background is described by combination of Gaussian and exponential functions. For considering the uncertainty in background shapes we replace single Gaussian function with double Gaussian function, allowed resolution to float and exponential function is replaced with 2^{nd} order Chebyshev

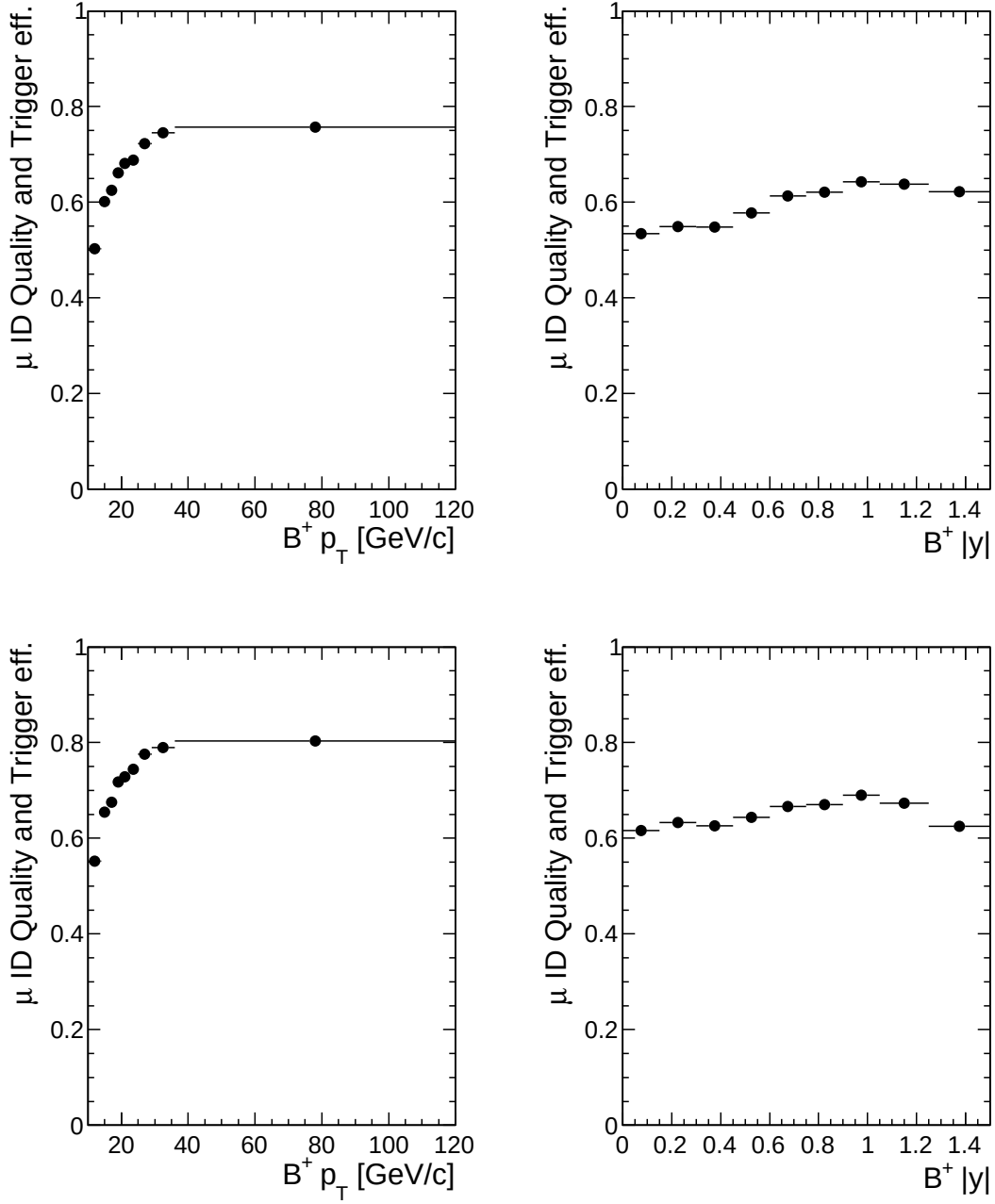


Figure 26: Dimuon efficiencies as a function of $B^\pm p_T$ (left) and $|y|$ (right). Upper plots are for the data and lower plots are for the MC samples.

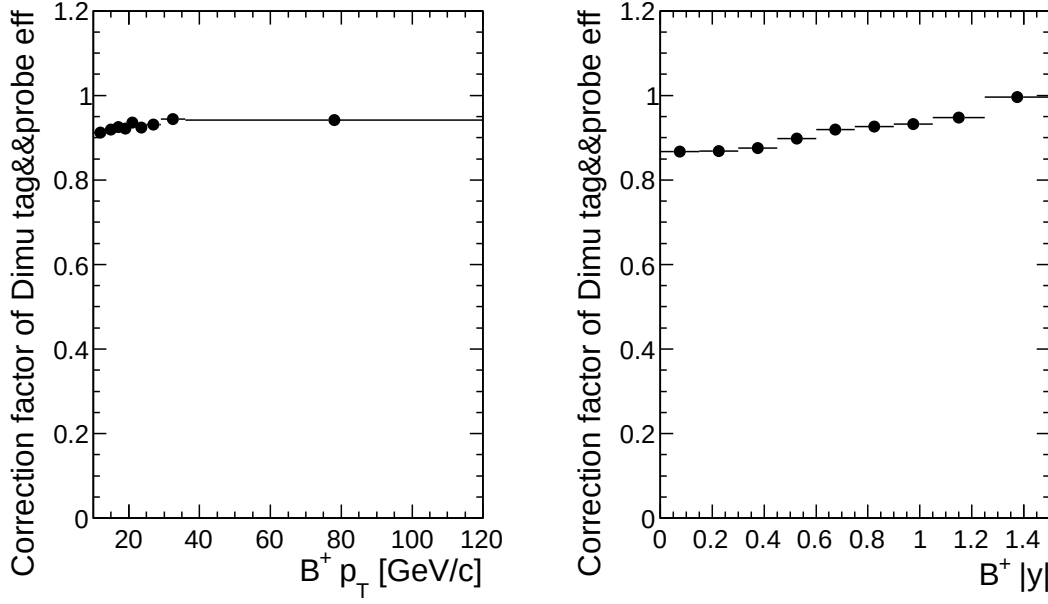


Figure 27: The correction factors of muon Tag&Probe efficiency $\alpha_{2\mu}$ for different $B^\pm p_T$ and $|y|$ bins.

polynomial. The difference in signal yield in each p_T and $|y|$ bin is quoted as signal shape and yields uncertainty.

4.3.3 Uncertainty on MC truth matching

We studied the effect of the ΔR matching and ends with a difference of 76 events. The ΔR with a cone size of 0.01 cut is based upon the J/ψ and π information. At first we extracted the yield without ΔR cut which is 16896. In the next step we applied the ΔR cut and extracted the yield again which comes out to be 16820. The left part of Figure 31 shows the J/ψ and π mass distribution with and without the MC truth matching. The distribution of the events failing the cut can be seen at the right of Figure 31.

Then we compare the signal yield between the method of MC truth matching with the ΔR cut and the method of mass spectrum fitting, the difference of the signal yield between two methods is the uncertainty of MC truth matching.

4.3.4 Uncertainty on energy scale

For differential cross sections measurements the uncertainty due to energy scale is a non-negligible effect because it will effect the number of $B_c(B^\pm)$ signal in each bin. Due to this uncertainty, an event can migrate within each differential bins. To compensate this effect we use the MC samples to compare the $B_c(B^\pm)$ fitting yield numbers and the $B_c(B^\pm)$ numbers in generator level using the truth matching technology in each p_T and $|y|$ bin, the difference in the $p_T(|y|)$ is quoted as uncertainty of energy scale.

4.3.5 Statistical uncertainty of Tag and Probe samples

It is the statistical uncertainties of Tag-Probe samples in the determination of muon related efficiencies. The single muon efficiency uncertainty is allowed to propagate to the $B_c(B^\pm)$ bins in the calculation of muon efficiency

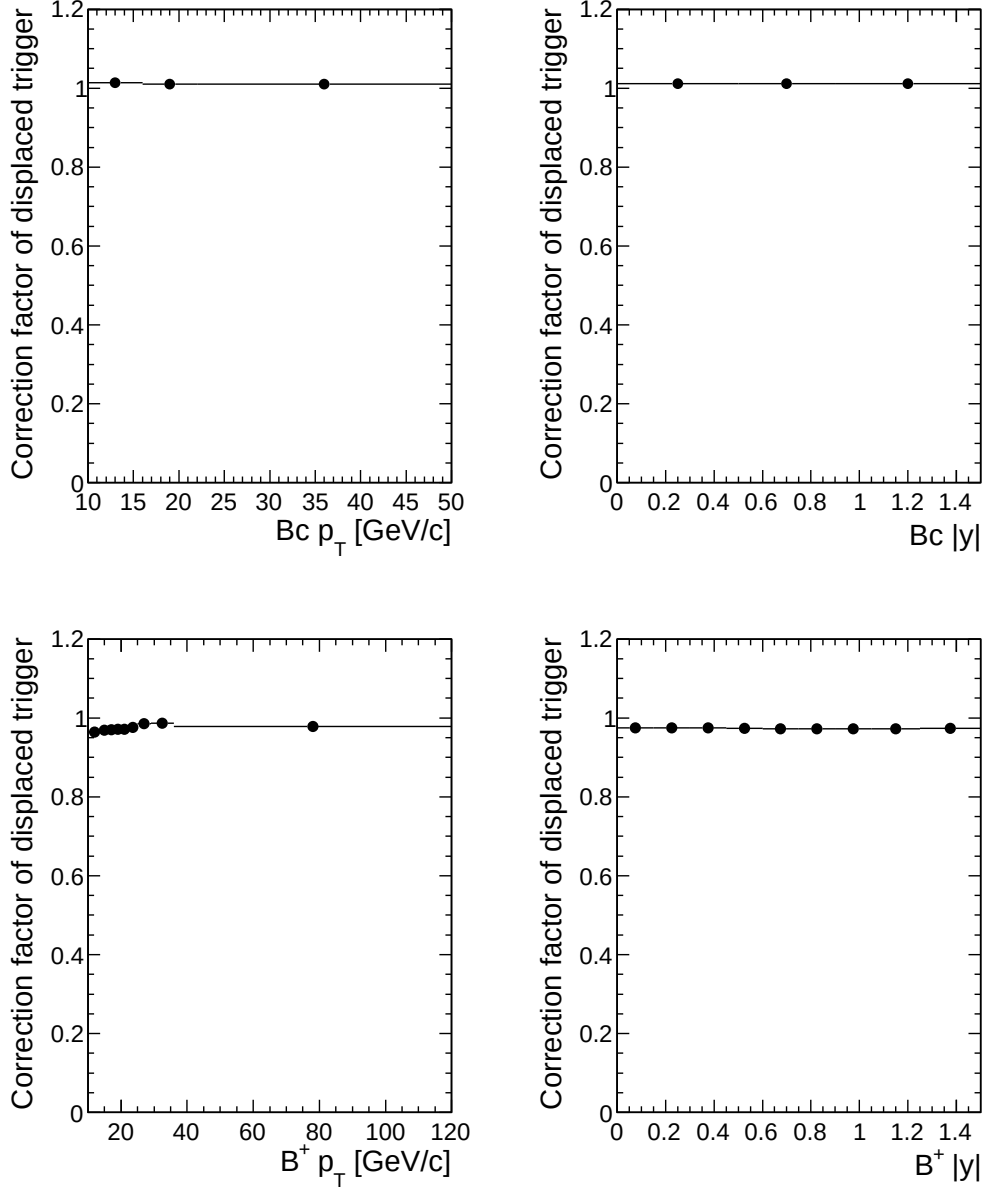


Figure 28: The correction factor α_{Disp} for different B_c (top) and $B^\pm$ (bottom) p_T and $|y|$ bins.

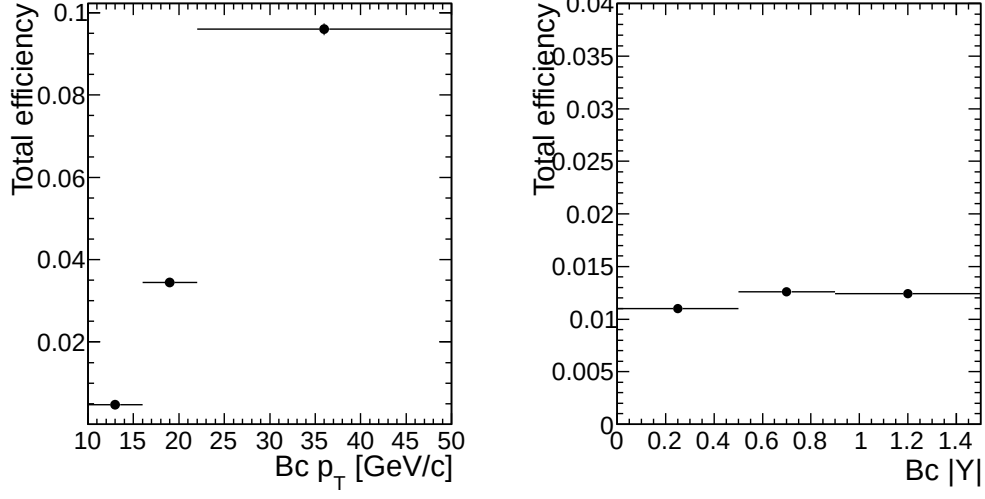


Figure 29: Total efficiency $\epsilon = Acc. \cdot \epsilon_{MC} \cdot \alpha_{2\mu} \cdot \alpha_{Disp}$, as a function of $B_c p_T$ (left) and rapidity (right).

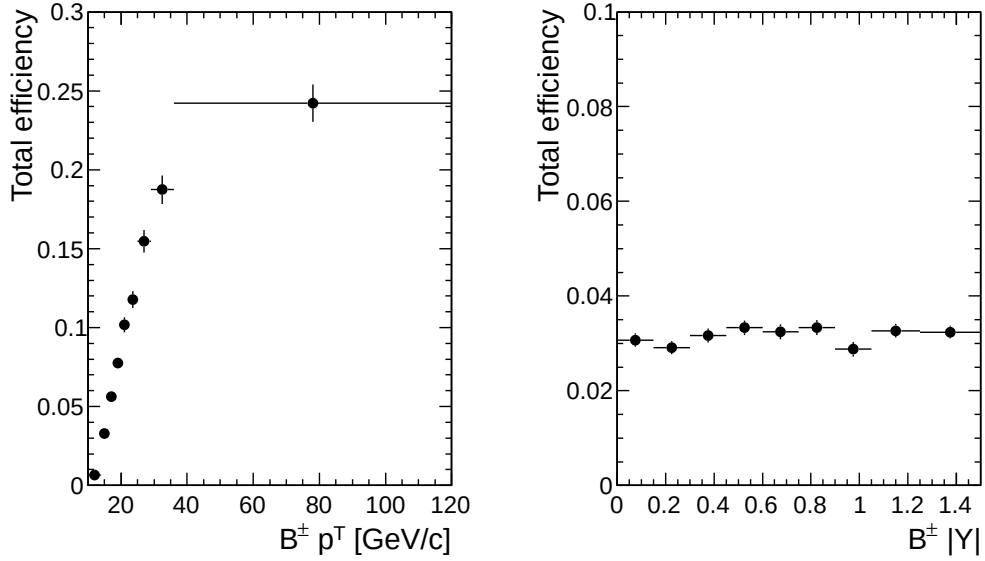


Figure 30: Total efficiency $\epsilon = Acc. \cdot \epsilon_{MC} \cdot \alpha_{2\mu} \cdot \alpha_{Disp}$, as a function of $B^\pm p_T$ (left) and rapidity (right).

Table 6: Displaced trigger efficiencies in J/ψ p_T and $|\eta|$ bins for different run periods in 2011. The uncertainties shown are statistical uncertainties in data.

Displaced trigger eff.	163269 – 167913	170249 – 172868	173236 – 180252	Average
$J\psi p_T : < 10.0 GeV$	0.955 ± 0.002	0.916 ± 0.006	0.861 ± 0.005	0.892 ± 0.003
$J\psi p_T : 10.0 - 12.0 GeV$	0.967 ± 0.002	0.926 ± 0.005	0.884 ± 0.004	0.911 ± 0.003
$J\psi p_T : 12.0 - 15.0 GeV$	0.974 ± 0.001	0.933 ± 0.005	0.880 ± 0.004	0.911 ± 0.003
$J\psi p_T : 15.0 - 20.0 GeV$	0.975 ± 0.002	0.929 ± 0.006	0.894 ± 0.005	0.919 ± 0.003
$J\psi p_T : 20.0 - 30.0 GeV$	0.978 ± 0.002	0.927 ± 0.008	0.899 ± 0.006	0.922 ± 0.004
$J\psi p_T : > 30.0 GeV$	0.979 ± 0.004	0.897 ± 0.017	0.910 ± 0.011	0.924 ± 0.007
$J\psi \eta : 0.0 - 0.4 GeV$	0.973 ± 0.002	0.938 ± 0.005	0.902 ± 0.004	0.925 ± 0.003
$J\psi \eta : 0.4 - 0.8 GeV$	0.970 ± 0.002	0.940 ± 0.005	0.898 ± 0.004	0.922 ± 0.003
$J\psi \eta : 0.8 - 1.2 GeV$	0.974 ± 0.002	0.930 ± 0.006	0.889 ± 0.005	0.916 ± 0.003
$J\psi \eta : 1.2 - 1.6 GeV$	0.966 ± 0.002	0.922 ± 0.006	0.862 ± 0.005	0.897 ± 0.003
$J\psi \eta : 1.6 - 2.2 GeV$	0.964 ± 0.002	0.896 ± 0.007	0.858 ± 0.006	0.889 ± 0.004

Table 7: Displaced trigger efficiencies in J/ψ p_T and $|\eta|$ bins for the B_c MC samples.

Displaced trigger eff.	Dimu7	Doublemu3p5	Doublemu4	Average
$J/\psi p_T : < 10.0 GeV$	0.915 ± 0.002	0.893 ± 0.004	0.862 ± 0.004	0.879 ± 0.003
$J/\psi p_T : 10.0 - 12.0 GeV$	0.934 ± 0.002	0.909 ± 0.003	0.883 ± 0.003	0.898 ± 0.002
$J/\psi p_T : 12.0 - 15.0 GeV$	0.940 ± 0.002	0.913 ± 0.003	0.889 ± 0.003	0.904 ± 0.002
$J/\psi p_T : 15.0 - 20.0 GeV$	0.943 ± 0.002	0.919 ± 0.003	0.895 ± 0.004	0.909 ± 0.002
$J/\psi p_T : 20.0 - 30.0 GeV$	0.948 ± 0.003	0.921 ± 0.005	0.896 ± 0.006	0.912 ± 0.004
$J/\psi p_T : > 30.0 GeV$	0.950 ± 0.007	0.927 ± 0.011	0.898 ± 0.013	0.914 ± 0.008
$J/\psi \eta : 0.0 - 0.4 GeV$	0.954 ± 0.002	0.934 ± 0.003	0.907 ± 0.003	0.922 ± 0.002
$J/\psi \eta : 0.4 - 0.8 GeV$	0.942 ± 0.002	0.920 ± 0.003	0.894 ± 0.003	0.909 ± 0.002
$J/\psi \eta : 0.8 - 1.2 GeV$	0.927 ± 0.002	0.900 ± 0.003	0.874 ± 0.004	0.890 ± 0.002
$J/\psi \eta : 1.2 - 1.6 GeV$	0.927 ± 0.002	0.899 ± 0.004	0.872 ± 0.004	0.888 ± 0.003
$J/\psi \eta : 1.6 - 2.2 GeV$	0.919 ± 0.003	0.895 ± 0.004	0.869 ± 0.005	0.884 ± 0.003

in Section 4.2.3.

4.3.6 Uncertainty on hadron tracking efficiency

The uncertainty on efficiency for reconstruction of a hadron track is 3.9% as in [54].

4.3.7 Uncertainty on muons kinematics

Because of limited statistics in muons efficiencies determination by the Tag and Probe technique, the average value of muon efficiency in each muon p_T and $|y|$ bin is used. To estimate the efficiency variance due to the muon kinematic distribution difference between the muon samples of Tag and Probe analysis and MC $B_c(B^\pm)$ sample, we re-weight the MC muon p_T distribution according to the data. Variance of efficiency determined by the re-weighted sample is considered to be uncertainty of muon kinematic distribution.

Table 8: Displaced trigger efficiencies in J/ψ p_T and $|\eta|$ bins for the $B^\pm$ MC samples.

Displaced trigger eff.	Dimu7	Doublemu3p5	Doublemu4	Average
$J\psi p_T : < 10.0 GeV$	0.966 ± 0.004	0.938 ± 0.007	0.915 ± 0.009	0.930 ± 0.005
$J\psi p_T : 10.0 - 12.0 GeV$	0.976 ± 0.003	0.950 ± 0.006	0.925 ± 0.007	0.940 ± 0.004
$J\psi p_T : 12.0 - 15.0 GeV$	0.976 ± 0.003	0.947 ± 0.005	0.922 ± 0.007	0.937 ± 0.004
$J\psi p_T : 15.0 - 20.0 GeV$	0.981 ± 0.003	0.952 ± 0.006	0.931 ± 0.007	0.945 ± 0.004
$J\psi p_T : 20.0 - 30.0 GeV$	0.970 ± 0.005	0.929 ± 0.009	0.905 ± 0.011	0.923 ± 0.007
$J\psi p_T : > 30.0 GeV$	0.987 ± 0.006	0.969 ± 0.013	0.946 ± 0.017	0.958 ± 0.010
$J\psi \eta : 0.0 - 0.4 GeV$	0.980 ± 0.003	0.956 ± 0.006	0.937 ± 0.007	0.949 ± 0.004
$J\psi \eta : 0.4 - 0.8 GeV$	0.968 ± 0.003	0.948 ± 0.006	0.928 ± 0.007	0.940 ± 0.004
$J\psi \eta : 0.8 - 1.2 GeV$	0.974 ± 0.003	0.943 ± 0.006	0.920 ± 0.008	0.936 ± 0.005
$J\psi \eta : 1.2 - 1.6 GeV$	0.974 ± 0.003	0.934 ± 0.007	0.900 ± 0.009	0.921 ± 0.005
$J\psi \eta : 1.6 - 2.2 GeV$	0.979 ± 0.003	0.947 ± 0.006	0.921 ± 0.008	0.937 ± 0.005

Table 9: Summary of the efficiencies measurements for different B_c p_T and $|y|$ bins. The uncertainties are statistical uncertainties of muon Tag&Probe sample and B_c MC samples.

$p_T^{B_c}$	Acc.	$\alpha_{2\mu}$	α_{Disp}	ε_{B_c-MC}	Total
10-16	0.1860 ± 0.0007	0.9033 ± 0.0001	1.0135 ± 0.0016	0.0276 ± 0.0004	0.00471 ± 0.00007
16-22	0.3663 ± 0.0022	0.9196 ± 0.0003	1.0104 ± 0.0014	0.1011 ± 0.0012	0.03441 ± 0.00047
22-50	0.5393 ± 0.0039	0.9312 ± 0.0004	1.0108 ± 0.0015	0.1890 ± 0.0023	0.09596 ± 0.00137
$ y^{B_c} $	Acc.	$\alpha_{2\mu}$	α_{Disp}	ε_{B_c-MC}	Total
0-0.5	0.1977 ± 0.0011	0.8584 ± 0.0002	1.0115 ± 0.0014	0.0641 ± 0.0008	0.01099 ± 0.00015
0.5-0.9	0.2056 ± 0.0012	0.9093 ± 0.0002	1.0115 ± 0.0016	0.0666 ± 0.0009	0.01259 ± 0.00019
0.9-1.5	0.2586 ± 0.0011	0.9518 ± 0.0002	1.0116 ± 0.0014	0.0499 ± 0.0006	0.01243 ± 0.00017

4.3.8 Uncertainty on B_c life time

The life time of $B_c = 0.452 \pm 0.041 ps$ [53]. We re-weighted our MC samples for B_c life time to be $0.411 ps$ and $0.493 ps$ at generator level(before any pre-selection). The difference in efficiencies of B_c after all selection criteria in both different B_c life time samples is considered to be uncertainty on measurements due to B_c life time uncertainty.

4.3.9 Uncertainty on muon tracking

Uncertainty on muon tracking taken to be 0.5% [54].

4.3.10 Uncertainty on other reconstruction efficiency

The J/ψ mass cut efficiency is very high $> 99.9\%$, the data and MC samples difference is less than 0.1% which is negligible. The $B_c(B^\pm)$ reconstruction efficiency determined from MC samples, we used the MC statistical uncertainty account for the the systematics uncertainty. For the systematic uncertainty of $B_c(B^\pm)$ vertex fit requirement, we compare the change of the data and MC efficiency with and without the $B_c(B^\pm)$ vertex fit requirement and difference is considered to be systematic uncertainty.

Table 10: Summary of the efficiencies measurements for different $B^\pm$ p_T and $|y|$ bins. The uncertainties are statistical uncertainties of muon Tag&Probe sample and $B^\pm$ MC samples.

$p_T^{B^\pm}$	Acc.	$\alpha_{2\mu}$	α_{Disp}	ε_{Bc-MC}	Total
10-14	0.2140 ± 0.0011	0.9116 ± 0.0006	0.9635 ± 0.0082	0.0356 ± 0.0015	0.00670 ± 0.00029
14-16	0.3343 ± 0.0029	0.9195 ± 0.0010	0.9687 ± 0.0076	0.1100 ± 0.0045	0.03277 ± 0.00138
16-18	0.3983 ± 0.0038	0.9250 ± 0.0011	0.9705 ± 0.0074	0.1574 ± 0.0062	0.05630 ± 0.00231
18-20	0.4582 ± 0.0049	0.9220 ± 0.0013	0.9714 ± 0.0080	0.1888 ± 0.0079	0.07748 ± 0.00343
20-22	0.5220 ± 0.0061	0.9357 ± 0.0015	0.9716 ± 0.0085	0.2145 ± 0.0095	0.10179 ± 0.00477
22-25	0.5548 ± 0.0063	0.9244 ± 0.0017	0.9763 ± 0.0075	0.2353 ± 0.0100	0.11781 ± 0.00527
25-29	0.6098 ± 0.0076	0.9312 ± 0.0020	0.9851 ± 0.0063	0.2765 ± 0.0120	0.15466 ± 0.00705
29-36	0.6672 ± 0.0085	0.9441 ± 0.0023	0.9861 ± 0.0065	0.3017 ± 0.0137	0.18739 ± 0.00892
36-120	0.7692 ± 0.0089	0.9412 ± 0.0034	0.9778 ± 0.0086	0.3421 ± 0.0160	0.24219 ± 0.01191
$ y^{B^\pm} $	Acc.	$\alpha_{2\mu}$	α_{Disp}	ε_{Bc-MC}	Total
0-0.15	0.2746 ± 0.0030	0.8676 ± 0.0013	0.9743 ± 0.0079	0.1322 ± 0.0060	0.03069 ± 0.00145
0.15-0.3	0.2738 ± 0.0030	0.8682 ± 0.0014	0.9749 ± 0.0079	0.1256 ± 0.0058	0.02910 ± 0.00141
0.3-0.45	0.2750 ± 0.0030	0.8754 ± 0.0012	0.9744 ± 0.0078	0.1351 ± 0.0061	0.03170 ± 0.00149
0.45-0.6	0.2674 ± 0.0030	0.8975 ± 0.0012	0.9739 ± 0.0077	0.1426 ± 0.0063	0.03332 ± 0.00153
0.6-0.75	0.2706 ± 0.0030	0.9192 ± 0.0012	0.9727 ± 0.0082	0.1342 ± 0.0062	0.03248 ± 0.00157
0.75-0.9	0.2795 ± 0.0031	0.9258 ± 0.0011	0.9729 ± 0.0083	0.1325 ± 0.0061	0.03335 ± 0.00161
0.9-1.05	0.3007 ± 0.0032	0.9321 ± 0.0013	0.9721 ± 0.0090	0.1056 ± 0.0053	0.02878 ± 0.00151
1.05-1.25	0.3355 ± 0.0029	0.9477 ± 0.0012	0.9727 ± 0.0074	0.1057 ± 0.0045	0.03268 ± 0.00144
1.25-1.5	0.3631 ± 0.0027	0.9955 ± 0.0010	0.9732 ± 0.0069	0.0921 ± 0.0037	0.03238 ± 0.00133

4.3.11 Uncertainty on alignment

We apply a systematic of 1% in this measurement to account for the effect due to imprecisely known alignment in CMS analysis $B^0 \rightarrow J/\psi K_s^0$ [55].

4.3.12 Uncertainty on branching fraction of $J/\psi \rightarrow \mu^+ \mu^-$

Uncertainties are taken from PDG [52] value for the J/ψ decay.

4.3.13 Uncertainty on luminosity estimation

We use an uncertainty of 4% on the luminosity estimate [48].

4.3.14 Split samples in high and low PU events

The data have been split in N sub-samples according to the number of Pile-Up vertices to check the consistency of the results and possible systematic effects.

In order to investigate the possible dependence of $B_c(B^+)$ cross section measurements on the Pile-Up, the efficiency behavior as a function of number of Primary Vertices has been studied. The MC samples have been divided in a low Pile-Up region, consisting of $1 \leq PV \leq 6$, and in a high Pile-Up region consisting of $PV \geq 7$. The efficiencies for the two regions are shown in Figure 32 for the $B_c \rightarrow J/\psi(\mu^+ \mu^-)\pi$ and Figure 33 $B^\pm \rightarrow J/\psi(\mu^+ \mu^-)K^\pm$ signals respectively.

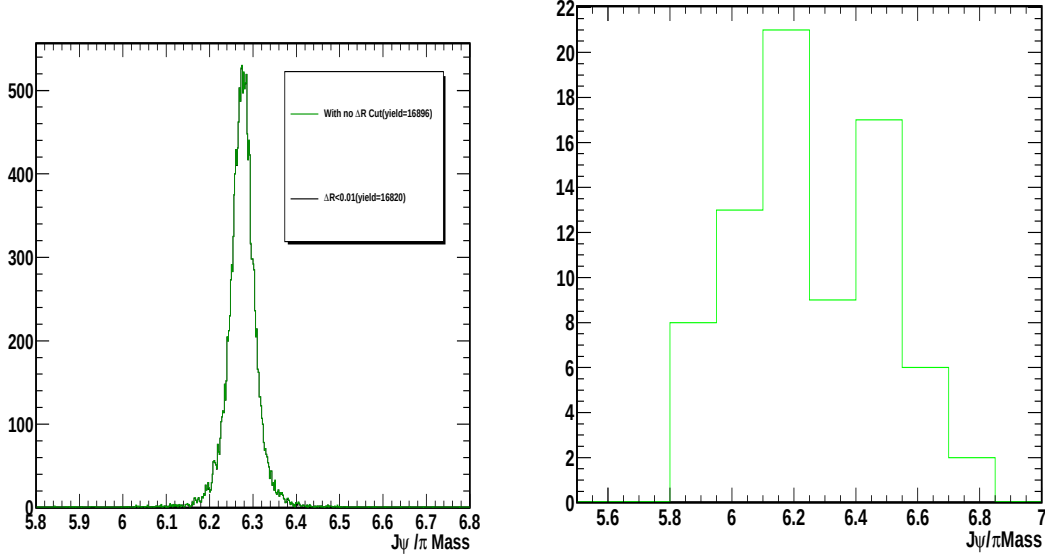


Figure 31: Left: Invariant mass distribution of the events with passing all cuts(black) and the events without ΔR cut(Green). Right: Mass distribution of the events failing the ΔR requirement

Subsequently, data have been divided according to the number of Primary Vertices in the event and signal events of $B_c(B^+)$ have been extracted separately for low and high Pile-Up region shows in Figure 34 and 35. The cross section has been evaluated in two regions as summarized in Table . The split samples turned out to be statistically consistent ($\chi^2 = 0.1$) for B_c and no additional uncertainty on B_c cross sections measurement has to be accounted for due to the PU dependence. Similar procedure has been used for B^+ as well and PU systatics has been reported in Table 12.

The resulting systematic uncertainties on the cross sections measurements of B_c and ($B^\pm$) are summarized in Table 11 and Table 12 respectively, where the total systematic uncertainties in each bin is calculated as the sum in quadrature of the individual uncertainties.

The uncertainties due to the muon Tag&Probe sample, $B_c(B^+)$ MC samples are not included in the Table 11 and Table 12. They are recorded in the efficiency measurement in Table 9 and Table 10.

4.4 Cross Sections Results

Using the number of observed $B_c(B^\pm)$ candidates and the reconstruction efficiencies, we get the values for the differential cross-sections of $B_c(B^\pm)$ times relative decay branching ratio.

At present, the predicted value of the branching ratio $\mathcal{B}(B_c \rightarrow J/\psi\pi)$ involves the non-perturbative transition form factors, which is of large errors [57–61]. For definiteness, we set the tree-level prediction $\mathcal{B}(B_c \rightarrow J/\psi\pi) = (1.3 \pm 0.8) \times 10^{-3}$ [57] as its default value. The differential cross sections of $d\sigma(B_c^+) \mathcal{B}(B_c \rightarrow J/\psi\pi)/dp_T^{B_c^+}$ and $d\sigma(B_c^+) \mathcal{B}(B_c^+ \rightarrow J/\psi\pi)/d|y|^{B_c^+}$ are listed in Table 13. As a comparison, the results predicted by BCVEGPY are also presented. Figure 42 shows the differential cross sections of $d\sigma(B_c^+) \mathcal{B}(B_c^+ \rightarrow J/\psi\pi^+)/dp_T^{B_c^+}$ and $d\sigma(B_c^+) \mathcal{B}(B_c^+ \rightarrow J/\psi\pi^+)/d|y|^{B_c^+}$ at $\sqrt{s} = 7$ TeV. The integrated production cross section for $B_c^+ \rightarrow J/\psi\pi^+$ with $p_T(B_c^+) > 10$ GeV and $|y(B_c^+)| < 1.5$ is $45.17 \pm 5.25_{-3.12}^{+3.00}$ pb, where the first uncertainty is statistical uncertainty and the second one is systematical uncertainty. Corresponding theoretical prediction for $\sigma(B_c^+) \mathcal{B}(B_c^+ \rightarrow J/\psi\pi^+)$ is 5.70 ± 3.51 pb, which used BCVEGPY production cross section $\sigma(B_c^+)$ and the branching ratio $\mathcal{B}(B_c \rightarrow$

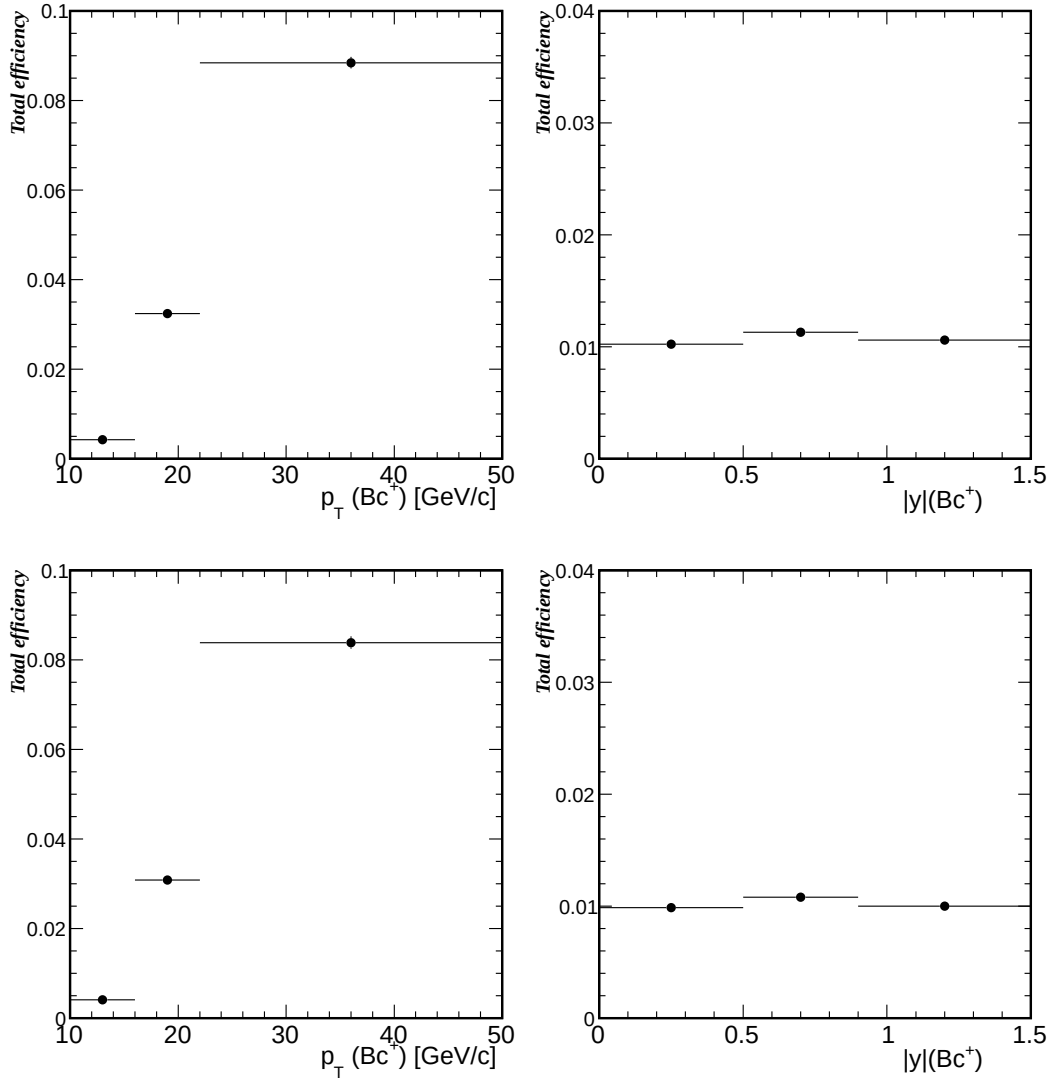


Figure 32: Total efficiency for B_c in low PU (top) and high PU (bottom) regions for different p_T and $|y|$ bins.

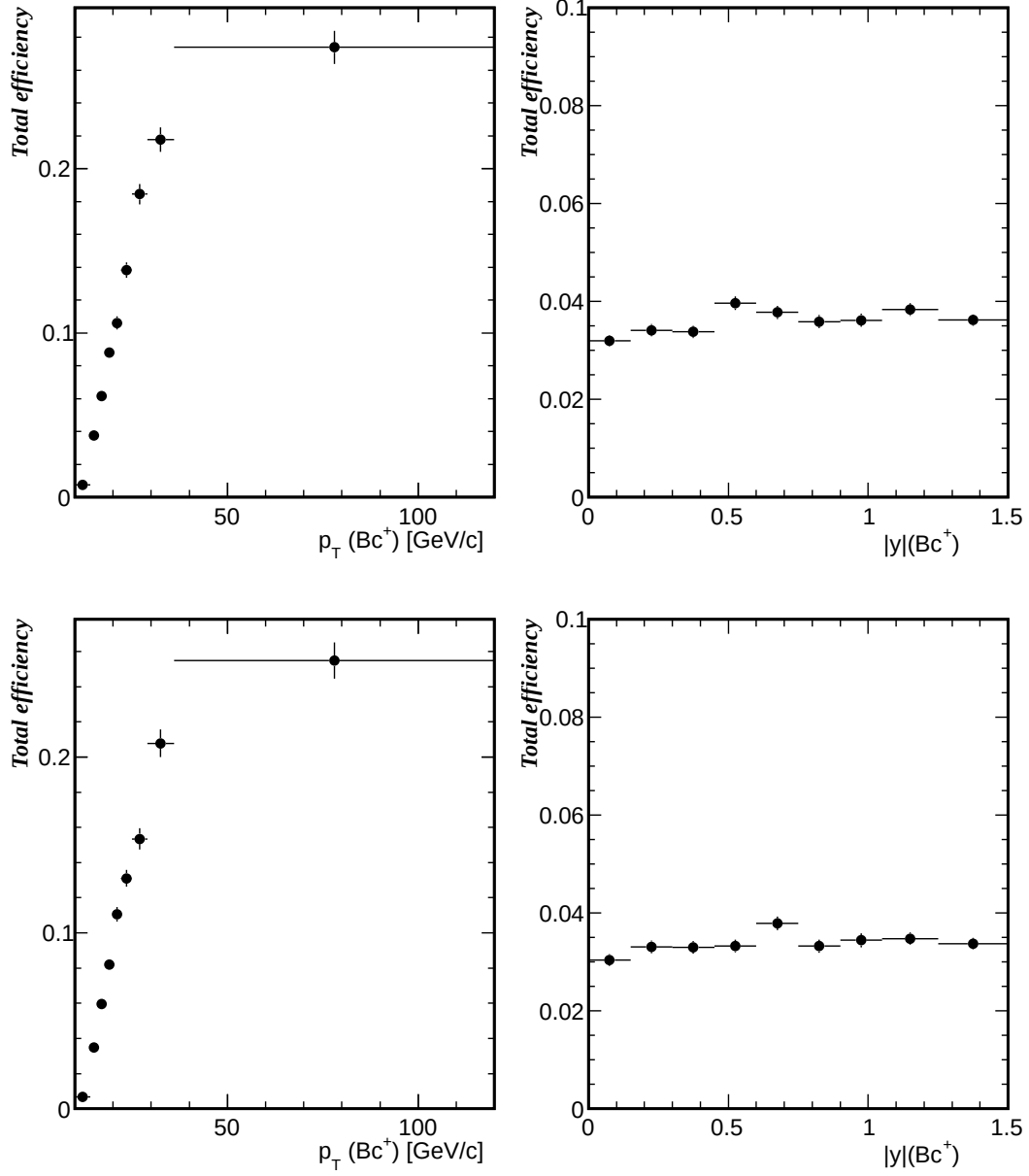


Figure 33: Total efficiency for $B^\pm$ in low PU (top) and high PU (bottom) regions for different p_T and $|y|$ bins.

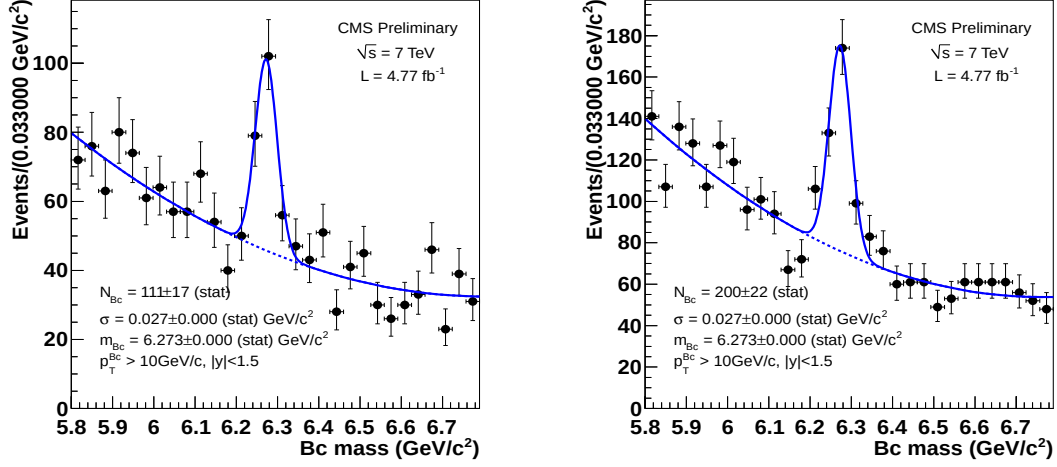
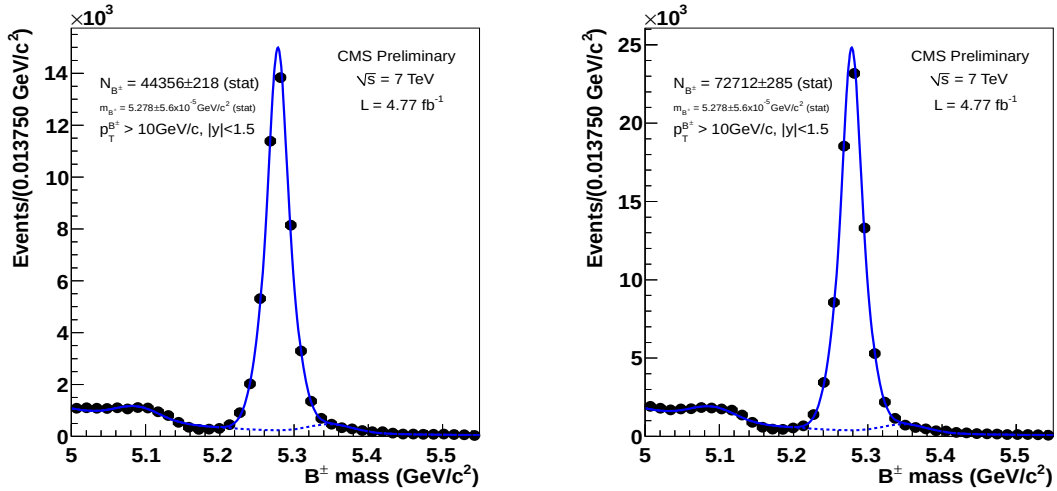
Figure 34: Inv. mass distribution of B_c for low PU (left) and high PU (right) regionsFigure 35: Inv. mass distribution of $B^\pm$ for low PU (left) and high PU (right) regions

Table 11: Uncertainties (in %) for B_c p_T and $|y|$ bins for the cross section measurement. The systematic uncertainties due to the muon Tap&Probe sample, B_c , B^+ MC samples are not included in the Table.

Source	p_T			$ y $			single
	Bin1	Bin2	Bin3	Bin1	Bin2	Bin3	Bin
Signal shape	0.8	6.4	0.9	0.5	0.7	0.3	2.4
Bg. shape	0.4	0.9	1.4	0.8	0.9	0.3	0.2
Truth Matching	0.1	0.3	0.2	0.4	0.2	0.2	0.3
Energy scale	0.6	0.2	0.3	0.1	0.1	0.1	0.1
Hadron tracking Efficiency	3.9	3.9	3.9	3.9	3.9	3.9	3.9
Muon kinematic	0.9	0.6	0.1	1.5	1.3	0.7	0.1
Muon Tracking	0.5	0.5	0.5	0.5	0.5	0.5	0.5
recon. eff.	0.4	0.3	0.5	0.4	0.3	0.4	0.2
Misalignment	1.0	1.0	1.0	1.0	1.0	1.0	1.0
B_c life time	+2.0 -3.4	+2.1 -2.7	+1.6 -2.2	+2.0 -2.6	+1.8 -2.9	+2.3 -2.8	+1.9 -2.8
$Br(J/\psi \rightarrow \mu^+ \mu^-)$	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Luminosity	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Total	+6.3 -6.9	+9.0 -9.1	+6.3 -6.4	+6.4 -6.6	+6.3 -6.7	+6.3 -6.5	+6.6 -6.9

$J/\psi\pi) = (1.3 \pm 0.8) \times 10^{-3}$ [we call this prediction as BCVEGPY prediction for simplicity]. Thus, there is large discrepancy between the BCVEGPY prediction and the measured one.

This discrepancy could be explained by the fact that I) The generator BCVEGPY is based on tree-level calculation on the gluon-gluon fusion production mechanism, and only the color-singlet $1S$ -wave and $1P$ -wave B_c states are considered. There are large uncertainties from the choices of the quark masses and the renormalization and factorization scales [49, 50, 62, 63]. The QCD corrections may also provide sizable contributions to the production cross section. II) The branching ratio $\mathcal{B}(B_c \rightarrow J/\psi\pi)$ involves the non-perturbative $B_c \rightarrow J/\psi X$ transition form factors. The QCD corrections to those form factors are large, cf. Refs. [64, 65]. It is also found that a correct choice of renormalization scale can achieve better predictions. For example, it has been found that by using the optimal scale determined by using principal of maximum conformality [66], one can get better B_c semi-leptonic decay width which agrees with the measurements [65]. As a prediction, if setting $\mathcal{B}(B_c \rightarrow J/\psi\pi) = 8 \times (1.3 \times 10^{-03})$, the cross section $\sigma(B_c)\mathcal{B}(B_c \rightarrow J/\psi\pi)$ shall show good agreement with our present measurement. Thus our present first measurement of B_c distributions stimulate a more detailed calculation on those input parameters.

Measurement for the differential cross sections of $d\sigma(B^+)\mathcal{B}(B^+ \rightarrow J/\psi K^+)/dp_T^{B^+}$ and $d\sigma(B^+)\mathcal{B}(B^+ \rightarrow J/\psi K^+)/d|y|^{B^+}$ has also been done. Corresponding efficiency and cross section results are listed in Table 14 for differential p_T and $|y|$ bins. The integrated production cross sections for $\sigma(B^+)\mathcal{B}(B^+ \rightarrow J/\psi K^+)$ with $p_T(B^+) > 10$ GeV and $|y(B^+)| < 1.5$ is measured to be $5919.02 \pm 82.33 \pm 426.17$ pb. The first uncertainty is statistical uncertainty and second is systematical uncertainty. Corresponding theoretical predictions by PYTHIA are also listed in the Table 14. 2D differential cross section for $\sigma(B^+)\mathcal{B}(B^+ \rightarrow J/\psi K^+)$ is also being reported for $0.0 < |y| < 0.75$ and $0.75 < |y| < 1.5$ for p_T bins 10-15, 15-18, 18-22, 22-28 and 28-120 GeV/c. The upper part of the Figure 44 explains the 2D differential cross sections measurement for differential $p_T(B^\pm)$ bins for $0.0 < |y| < 0.75$ at left and $0.75 < |y| < 1.5$ at right. The bottom of the Figure 44 is $B^+ \rightarrow J/\psi K^+$ cross sections measurement for differential p_T bins at left bottom and right bottom is the measurement results in differential $|y|$ bins. In case of B^+ , the integrated cross sections are consistent within one sigma between the experimental measurement and the PYTHIA prediction. A decent comparison of the measurement with other theoretical approaches(FONLL and NLO) has also been made. In case of FONLL and NLO the calculations has

been performed assuming hadronization fraction of $f_{b\bar{b} \rightarrow B^+} = 0.337 \pm 0.002$. In case of FONLL the measured cross section is $4896.82^{+1667.58}_{-1086.66}(\text{scale}) \pm 249.3(\text{mass})$ pb. The later results are quoted with the theoretical uncertainty in the b-quark mass, renormalization and factorization scales. For FONLL, the default calculations the mass and the scales used are set to 4.75 ± 0.25 GeV/ c^2 and $\mu_F = \mu_R = \mu_0$ [56]. Uncertainties from renormalization and factorization scales are estimated by varying them up and down by a factor of two ($\mu_0/2 < \mu_R$, with $\mu_F < 2\mu_0/2 < \mu_R/\mu_F < 2$) [56]. In estimating the b-quark mass uncertainty, the mass has been changed from 4.75 GeV/ c^2 to 5.00 GeV/ c^2 and 4.5 GeV/ c^2 [56]. The theoretical uncertainty is quite large (35%). The theoretical uncertainty of NLO and PYTHIA are similar to FONLL. It is shown that the measurements are in good agreement to the theoretical predictions within the theoretical uncertainty. The consistency of the B^+ event with high p_T and $|y|$ is better than the events with low p_T and $|y|$. The $B_c \rightarrow J/\psi\pi$ and $B^\pm \rightarrow J/\psi K$ cross section measurements has been cross checked by measuring the ratio $R = \frac{\sigma(B_c^+) \times BR(B_c \rightarrow J/\psi\pi)}{\sigma(B^+) \times BR(B^\pm \rightarrow J/\psi K)}$ and the results (0.0049 ± 0.0006) are found to be consistent with the one quoted in reference [41].

5 Summary

The $B_c^+ \rightarrow J/\psi\pi^+$, $B^+ \rightarrow J/\psi K^+$ production cross sections are measured at LHC/CMS. The dataset is collected by the CMS detector in 2011 at $\sqrt{s} = 7$ TeV, which corresponds to an integrated luminosity of 4.77 fb^{-1} . In the analysis the B_c ($B^\pm$) mesons are reconstructed by the exclusive decay $B_c^\pm \rightarrow J/\psi\pi^\pm$, $B^\pm \rightarrow J/\psi K^\pm$, and J/ψ is reconstructed by its dimuon decay. The B_c^+ , B^+ differential cross sections are reported with respect to $p_T(B_c^+, B^+)$ and rapidity $|y|$. The cross sections were measured as a function of transverse momentum and rapidity while requiring $p_T > 10 \text{ GeV}$ and $|y| < 1.5$. In case of $B_c \rightarrow J/\psi(\mu^+\mu^-)\pi$ the measurement has been compared with the theoretical predictions from BCVEGPY. The predictions are found to be lower from measurement by a factor of 8 with $\mathcal{B}(B_c \rightarrow J/\psi\pi) = 1.3 \times 10^{-03}$ while the shapes are in good agreement. The measurement of the cross sections for $B^+ \rightarrow J/\psi K^+$ has been compared with the predictions from PYTHIA. The predictions from the NLO and FONLL theoretical framework have also been quoted with the theoretical uncertainty from the renormalization and factorization scale and b-quark mass (35%). The consistency of the B^+ event with high p_T and $|y|$ is better than the events with low p_T and $|y|$. The measurement is found to be in good consistency with the predictions in FONLL theoretical framework within the theoretical uncertainty. We wish the measurements may be helpful for theorists to understand and investigate further the B_c and $B^\pm$ production and decay mechanism.

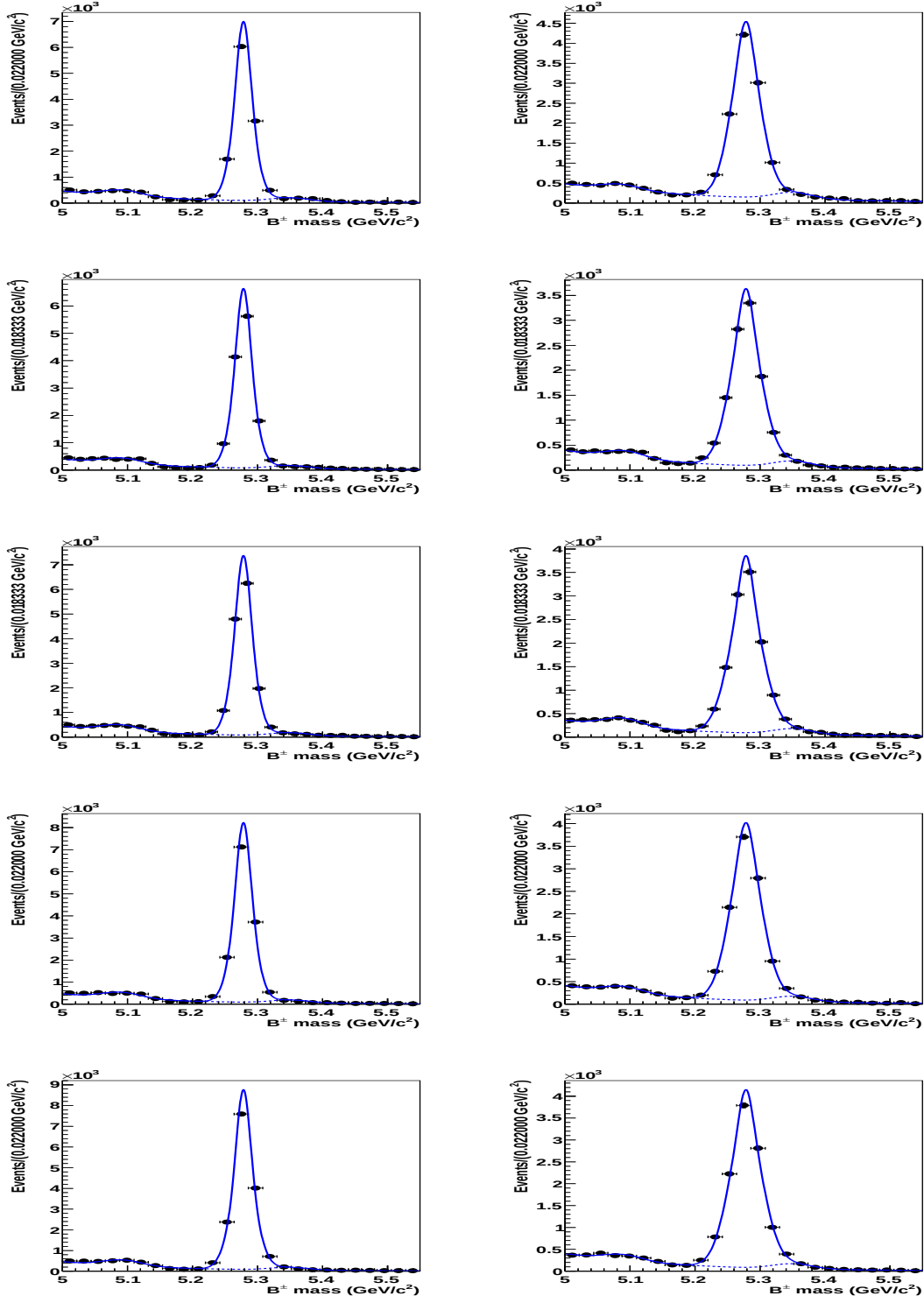


Figure 36: Mass fit plots of $J/\psi K^\pm$ 2D candidates. left: for small rapidity $0.0 < |y| < 0.75$, right: for large rapidity $0.75 < |y| < 1.5$, top to bottom is for the five p_T differential bins of $B^\pm$.

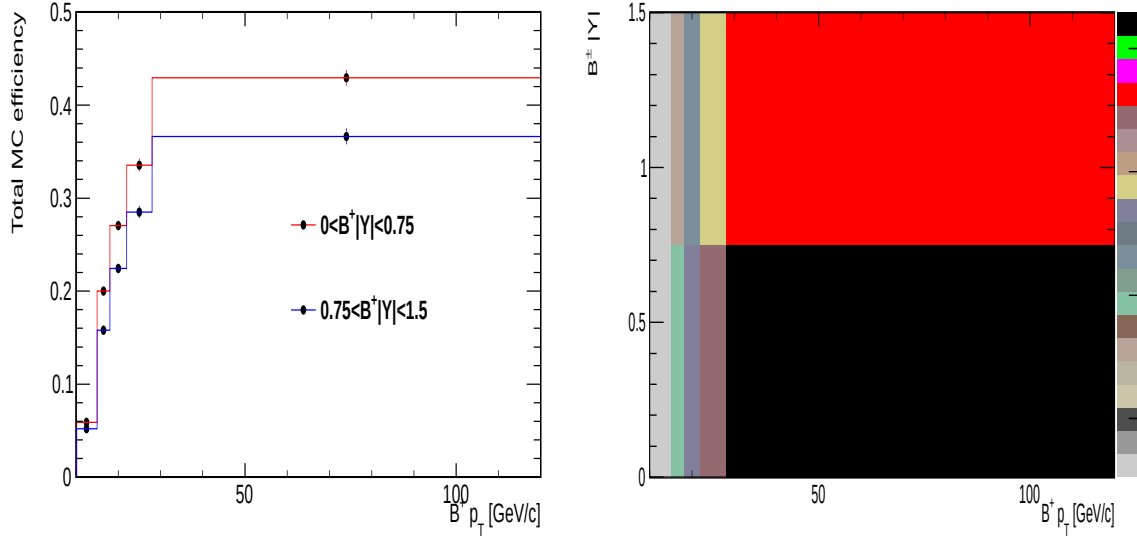


Figure 37: ϵ_{MC} of $B^+ \rightarrow J/\psi K^+$ for 2D candidates in different p_T and $|y|$ bins.

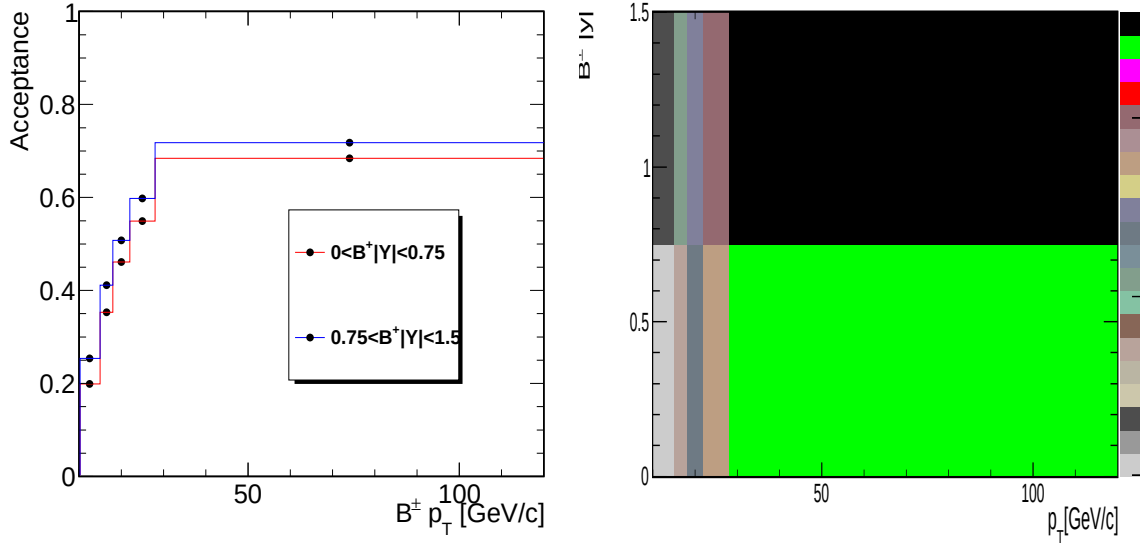


Figure 38: Acceptance of $B^+ \rightarrow J/\psi K^+$ for 2D candidates in different p_T and $|y|$ bins.

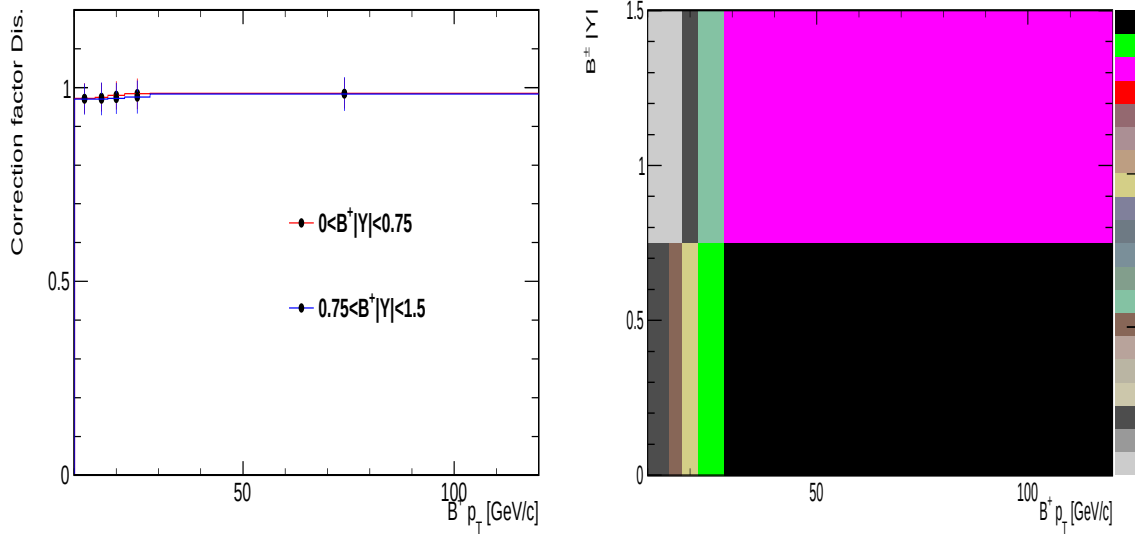


Figure 39: $\alpha_{Disp-trigger}$ of $B^+ \rightarrow J/\psi K^+$ for 2D candidates in different p_T and $|y|$ bins.

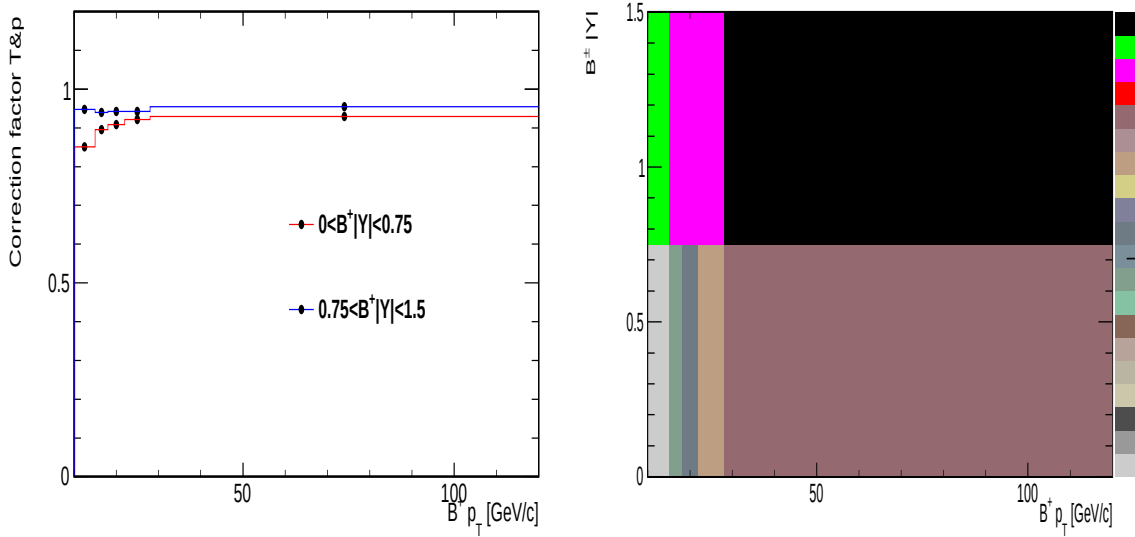


Figure 40: α_{T-P} of $B^+ \rightarrow J/\psi K^+$ for 2D candidates in different p_T and $|y|$ bins.

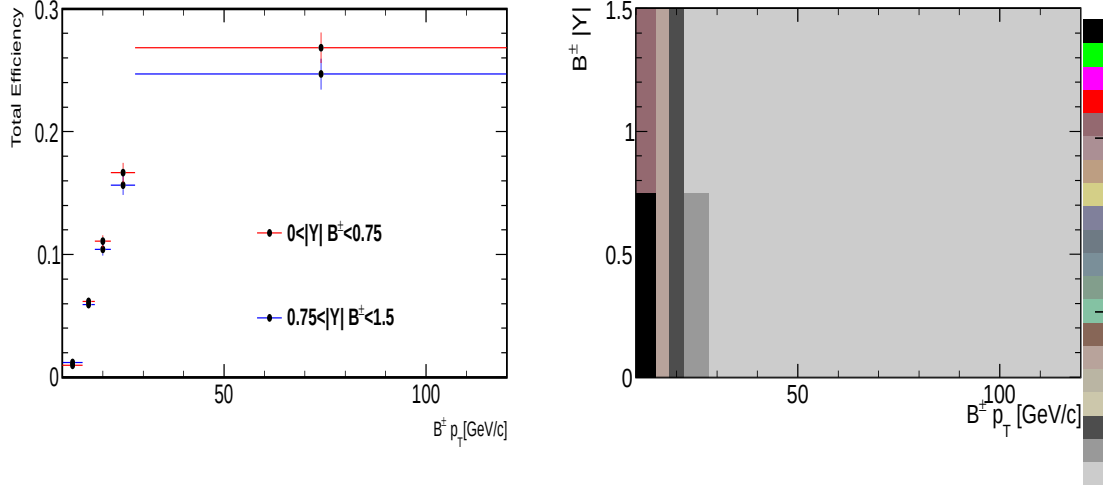


Figure 41: The 2D $\epsilon_{Tot.}$ of $B^+ \rightarrow J/\psi K^+$ for different p_T and $|y|$ bins.

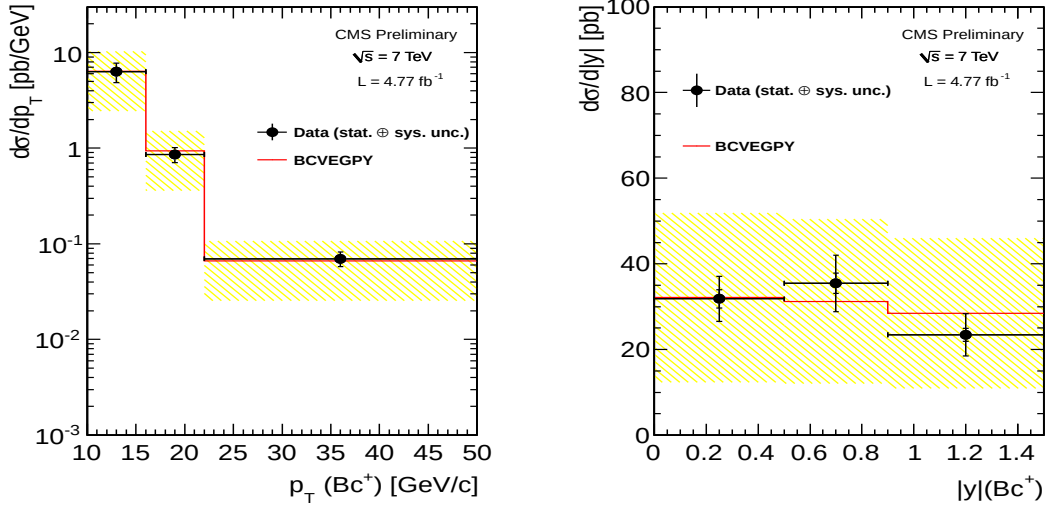


Figure 42: Differential production cross sections of $d\sigma/dp_T^{B_c^+ \rightarrow J/\psi \pi^+}$ and $d\sigma/d|y|^{B_c \rightarrow J/\psi \pi^+}$ at $\sqrt{s} = 7$ TeV pp collisions. Dots with uncertainties bars are the CMS measurements, uncertainties are systematical uncertainties and statistic uncertainties respectively. Red histogram is BCVEGPY prediction. The uncertainties of BCVEGPY is branching ratio uncertainties of the $Br(B_c \rightarrow J/\psi \pi)$ which is $1.3 \times 10^{-03} \pm 8.0 \times 10^{-04}$ in reference [57].

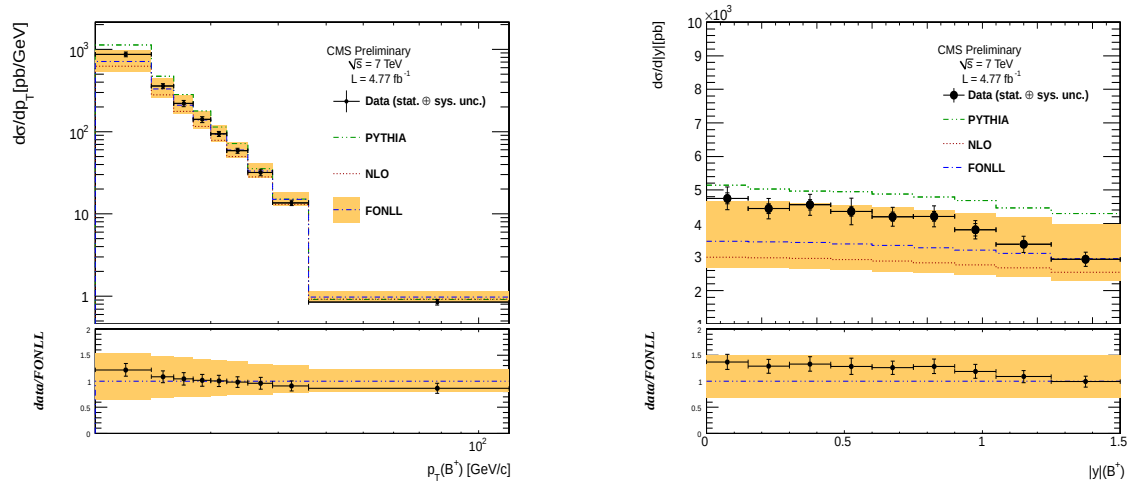


Figure 43: Differential production cross sections of $d\sigma/dp_T^{B^+ \rightarrow J/\psi K^+}$ and $d\sigma/d|y|^{B^+ \rightarrow J/\psi K^+}$ at $\sqrt{s} = 7$ TeV pp collisions. Dots with uncertainties are the CMS measurements, uncertainties are systematical uncertainties and statistical uncertainties respectively. The color bands are PYTHIA predictions. The uncertainties of PYTHIA are branching ratio uncertainties of the $Br(B^\pm \rightarrow J/\psi K^\pm)$ which is $1.014 \times 10^{-03} \pm 3.4 \times 10^{-05}$ in PDG [52].

Table 12: Uncertainties (in %) for B^+p_T and $|y|$ bins for the cross sections measurements.

source	p_T									$ y $									Single
	Bin1	Bin2	Bin3	Bin4	Bin5	Bin6	Bin7	Bin8	Bin9	Bin1	Bin2	Bin3	Bin4	Bin5	Bin6	Bin7	Bin8	Bin9	Bin
Signal shape	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Bg. shape	2.4	3.9	6.0	5.8	3.9	3.5	4.5	4.3	5.3	4.0	3.2	3.5	4.6	3.1	4.5	4.1	4.0	4.0	3.8
Truth Matching	0.2	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.1	0.4	0.4	0.4	0.2	0.2	0.2	0.2	0.2	0.2	0.3
Energy scale	0.6	0.6	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Hadron track Eff.	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9
Muon kinematic	0.9	0.9	0.6	0.6	0.6	0.1	0.1	0.1	0.1	1.5	1.5	1.5	1.3	1.3	1.3	0.7	0.7	0.7	0.1
Muon Tracking	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
recon. eff.	0.4	0.4	0.3	0.3	0.3	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.4	0.4	0.4	0.2
Misalignment	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
$Br(J/\psi \rightarrow \mu^+\mu^-)$	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Luminosity	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pileup	1.5	0.0	0.0	0.0	0.0	0.0	5.5	0.0	0.0	0.0	0.0	0.0	5.2	0.0	0.0	0.0	1.3	0.0	2.1
Total	6.6	7.1	8.4	8.2	7.0	6.8	9.2	7.2	7.9	7.2	6.8	6.9	9.1	6.7	7.5	7.1	7.2	7.1	6.9

Table 13: Measurements of differential cross sections $d\sigma/dp_T^{B_c^+ \rightarrow J/\psi\pi^+}$, $d\sigma/d|y|^{B_c^+ \rightarrow J/\psi\pi^+}$ and integrated cross section for B_c $p_T > 10$ GeV/ c and $|y| < 1.5$. The first uncertainties are statistical uncertainties of data and signal MC samples, the second are systematical uncertainties. The uncertainty of BCVEGPY is branching ratio uncertainty of $Br(B_c \rightarrow J/\psi\pi)$ which is $1.3 \times 10^{-03} \pm 8.0 \times 10^{-04}$ in reference [57].

$p_T^{B_c^+}$	n_{sig}	ϵ	$d\sigma/dp_T^{B_c^+}$ (pb/GeV/ c)	BCVEGPY
10-16	101.0 ± 23.0	0.00471 ± 0.00007	$6.29 \pm 1.44_{-0.43}^{+0.40}$	0.79 ± 0.49
16-22	101.3 ± 18.1	0.03441 ± 0.00047	$0.86 \pm 0.15_{-0.08}^{+0.08}$	0.12 ± 0.07
22-50	107.0 ± 18.1	0.09596 ± 0.00137	$0.07 \pm 0.01_{-0.01}^{+0.01}$	0.01 ± 0.01
$ y ^{B_c^+}$	n_{sig}	ϵ	$d\sigma/d y ^{B_c^+}$ (pb)	BCVEGPY
0-0.5	99.6 ± 16.3	0.01099 ± 0.00015	$31.82 \pm 5.23_{-2.10}^{+2.03}$	4.02 ± 2.47
0.5-0.9	101.6 ± 18.8	0.01259 ± 0.00019	$35.46 \pm 6.58_{-2.38}^{+2.23}$	3.90 ± 2.40
0.9-1.5	99.3 ± 20.9	0.01243 ± 0.00017	$23.39 \pm 4.95_{-1.52}^{+1.47}$	3.56 ± 2.19
n_{sig}		ϵ	σ (pb)	BCVEGPY
Single bin	309.68 ± 35.78	0.012041 ± 0.000097	$45.17 \pm 5.25_{-3.12}^{+3.00}$	5.70 ± 3.51

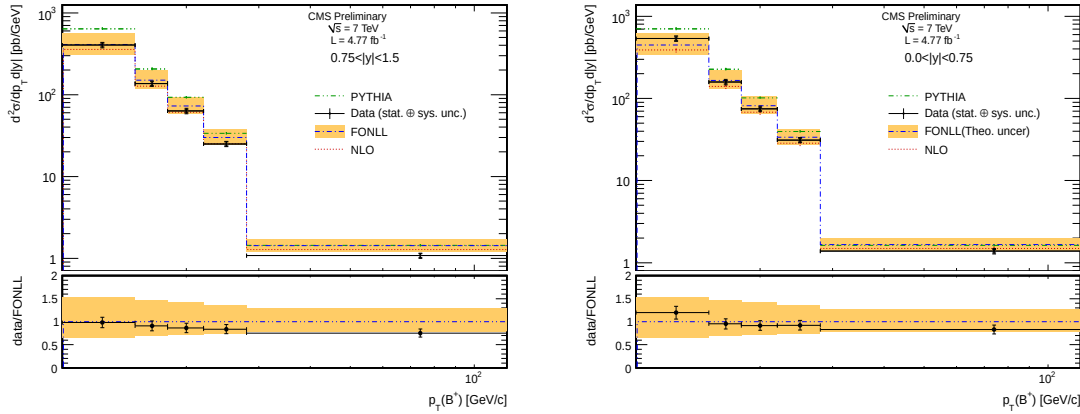


Figure 44: 2D ($p_T - |y|$) differential production cross sections for $B^+ \rightarrow J/\psi K^+$ at $\sqrt{s} = 7$ TeV pp collision left: for $0.0 < |y|^{B^+} < 0.75$, right: for $0.75 < |y|^{B^+} < 1.5$. Dots with uncertainties are the CMS measurement, uncertainties are systematics and statistics uncertainties respectively. The color bands covers the PYTHIA predictions whereas the width of bands involve the error arising due to the error of PYTHIA for $Br(B^\pm \rightarrow J/\psi K^\pm)$ of $1.014 \times 10^{-03} \pm 3.4 \times 10^{-05}$ in PDG [52]. The theoretical predictions of NLO and FONLL has also been added where the error is the fraction of $b\bar{b} \rightarrow B^\pm + X$ (0.337 ± 0.020) and of $Br(B^\pm \rightarrow J/\psi K^\pm)$ to be $1.014 \times 10^{-03} \pm 3.4 \times 10^{-05}$.

Table 14: Measurement of differential cross sections of $d\sigma/dp_T^{B^+ \rightarrow J/\psi K^+}$, $d\sigma/d|y|^{B^+ \rightarrow J/\psi K^+}$ and integrated cross section for $B^\pm$ $p_T > 10$ GeV/ c and $|y| < 1.5$. The first uncertainties are statistical uncertainties of data sample, signal MC samples, the second are systematical uncertainties. The uncertainties of PYTHIA is branching ration uncertainty of $Br(B^\pm \rightarrow J/\psi K^\pm)$ which is $1.014 \times 10^{-03} \pm 3.4 \times 10^{-05}$ in PDG.

$p_T^{B^+}$	n_{sig}	ε	$d\sigma/dp_T^{B^+}$ (pb GeV/ c)	PYTHIA $\pm$ BR Uncertainty
10-14	14277.3 ± 128.1	0.00722 ± 0.00018	$868.55 \pm 26.94 \pm 57.32$	1126.29 ± 37.82
14-16	14752.0 ± 129.7	0.03626 ± 0.00088	$357.32 \pm 10.84 \pm 23.58$	470.37 ± 16.00
16-18	15148.2 ± 121.5	0.06052 ± 0.00147	$219.82 \pm 6.56 \pm 14.51$	282.74 ± 9.72
18-20	13612.1 ± 118.7	0.08514 ± 0.00218	$140.42 \pm 4.42 \pm 9.27$	177.81 ± 6.20
20-22	11573.4 ± 111.8	0.10819 ± 0.00303	$93.95 \pm 3.21 \pm 6.20$	113.64 ± 4.05
22-25	13493.9 ± 120.9	0.13474 ± 0.00349	$58.64 \pm 1.87 \pm 3.87$	71.80 ± 2.57
25-29	12353.7 ± 110.3	0.16946 ± 0.00457	$32.01 \pm 1.05 \pm 2.11$	35.61 ± 1.31
29-36	11596.1 ± 111.3	0.21292 ± 0.00576	$13.67 \pm 0.46 \pm 0.90$	15.21 ± 0.58
36-120	10721.0 ± 109.3	0.26443 ± 0.00751	$0.85 \pm 0.03 \pm 0.06$	0.92 ± 0.04
$ y ^{B^+}$	n_{sig}	ε	$d\sigma/d y ^{B^+}$ (pb)	PYTHIA $\pm$ BR Uncertainty
0-0.15	12637.7 ± 113.2	0.03116 ± 0.00088	$4749.64 \pm 161.00 \pm 341.97$	5145.29 ± 175.58
0.15-0.3	12745.3 ± 116.4	0.03359 ± 0.00092	$4443.19 \pm 147.98 \pm 319.91$	5031.86 ± 171.79
0.3-0.45	12993.2 ± 116.5	0.03338 ± 0.00094	$4558.09 \pm 153.53 \pm 328.18$	4970.09 ± 169.72
0.45-0.6	13583.0 ± 119.0	0.03649 ± 0.00100	$4359.70 \pm 143.19 \pm 313.90$	4946.21 ± 168.92
0.6-0.75	13561.3 ± 120.8	0.03778 ± 0.00102	$4203.86 \pm 137.30 \pm 302.68$	4879.39 ± 166.68
0.75-0.9	12438.3 ± 117.6	0.03457 ± 0.00099	$4213.29 \pm 144.93 \pm 303.36$	4789.15 ± 163.66
0.9-1.05	11492.5 ± 247.0	0.03530 ± 0.00102	$3813.04 \pm 183.58 \pm 274.54$	4688.12 ± 160.28
1.05-1.25	14075.1 ± 126.3	0.03658 ± 0.00094	$3379.31 \pm 106.92 \pm 243.31$	4465.68 ± 151.99
1.25-1.5	14603.2 ± 133.6	0.03499 ± 0.00085	$2932.80 \pm 89.86 \pm 211.16$	4299.97 ± 145.94
	n_{sig}	ε	σ (pb)	PYTHIA $\pm$ BR Uncertainty
Single bin	117486.38 ± 359.44	0.03509 ± 0.00032	$5919.02 \pm 82.33 \pm 426.17$	7135.64 ± 239.30

Table 15: Separation of p_T (GeV/ c) and $|y|$ bins for B^+ 2D differential cross sections measurements.

$B^\pm$	p_T Bin1	p_T Bin2	p_T Bin3	p_T Bin4	p_T Bin5
$0 < y < 0.75$	10-15	15-18	18-22	22-28	> 28
$0.75 < y < 1.5$	10-15	15-18	18-22	22-28	> 28

Table 16: $B^\pm$ 2D efficiencies measurements.

p_T GeV/ c	ϵ_{MC}		$Acc.$		$\alpha_{Disp-trigger}$		α_{T-P}	
	$ y $		$ y $		$ y $		$ y $	
	0.0-0.75	0.75-1.5	0.0-0.75	0.75-1.5	0.0-0.75	0.75-1.5	0.0-0.75	0.75-1.5
10-15	0.059 ± 0.002	0.052 ± 0.001	0.199 ± 0.001	0.254 ± 0.002	0.972 ± 0.039	0.970 ± 0.040	0.851 ± 0.001	0.947 ± 0.000
15-18	0.200 ± 0.004	0.158 ± 0.004	0.353 ± 0.004	0.411 ± 0.004	0.975 ± 0.037	0.971 ± 0.042	0.896 ± 0.001	0.940 ± 0.001
18-22	0.270 ± 0.006	0.224 ± 0.005	0.461 ± 0.005	0.507 ± 0.006	0.980 ± 0.037	0.972 ± 0.040	0.908 ± 0.001	0.942 ± 0.001
22-28	0.335 ± 0.007	0.285 ± 0.007	0.549 ± 0.007	0.598 ± 0.007	0.984 ± 0.040	0.976 ± 0.043	0.921 ± 0.001	0.942 ± 0.001
28-120	0.429 ± 0.008	0.366 ± 0.008	0.684 ± 0.008	0.718 ± 0.008	0.985 ± 0.039	0.984 ± 0.044	0.930 ± 0.002	0.955 ± 0.001

Table 17: $B^\pm$ 2D cross sections measurements.

	$N_{B^\pm}$		$\epsilon_{Tot.}$		$d^2\sigma/(d y dp_T)$		PYTHIA	
p_T GeV/ c	$ y $		$ y $		$ y $		$ y $	
	0.0-0.75	0.75-1.5	0.0-0.75	0.75-1.5	0.0-0.75	0.75-1.5	0.0-0.75	0.75-1.5
10-15	11042.8 ± 108.33	10478.8 ± 111.64	0.0096 ± 0.0005	0.0122 ± 0.0006	536.051 ± 27.001	403.867 ± 20.689	704.019 ± 23.697	639.230 ± 21.528
15-18	12485.9 ± 113.92	10360.8 ± 111.1	0.062 ± 0.0028	0.059 ± 0.003	158.188 ± 7.455	136.677 ± 7.292	226.725 ± 7.813	206.243 ± 7.127
18-22	14096.4 ± 116.6	11229.4 ± 115.997	0.111 ± 0.0049	0.104 ± 0.005	74.510 ± 3.440	63.131 ± 3.204	101.859 ± 3.576	92.444 ± 3.261
22-28	13295.7 ± 117.82	10030 ± 105.87	0.167 ± 0.008	0.157 ± 0.008	31.153 ± 1.537	25.008 ± 1.334	39.407 ± 1.431	33.769 ± 1.239
28-120	14578.1 ± 123.29	10440.9 ± 108.78	0.268 ± 0.0123	0.247 ± 0.013	1.383 ± 0.066	1.076 ± 0.057	1.631 ± 0.062	1.439 ± 0.055

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